PS 1.

1. a) $n = a^k b$ for some $k \in \mathbb{Z}^+$

Show $\overline{ab}$ is nilpotent.

Consider $(ab)^k$, $(ab)^k = abab \dots ab = a^k b^k$ since integers commute

$\Rightarrow (ab)^k = a^k b^k = a^k b \cdot b^{k-1} = n b^{k-1}$

Now in $\mathbb{Z}/n\mathbb{Z}$, $nb^{k-1} \equiv 0 \pmod{n}$

$\Rightarrow (ab)^k = 0$ ~~for~~, and $k \in \mathbb{Z}^+ \Rightarrow (ab)$ nilpotent.

b). Suppose $a$ is nilpotent. $\exists k \in \mathbb{Z}^+ : \bar{a}^k = 0$

$\Rightarrow n | a^k$, let $p$ be a prime dividing $n$

$p | n$. ~~Now~~ Since $n | a^k$, then $p | a^k$.

But since $p$ is prime, $p | a$

Now suppose every prime divisor of $n$ is a divisor of $a$. Consider the prime factorization of $n$.

$n = \sum p_i^{k_i}$.

Let $m = \prod k_i$. ~~Every~~ Since every prime divisor of $n$ divides $a$, $a = \prod p_i \cdot b$, for some $b \in \mathbb{Z}^+$

Now, consider $a^m$, $a^m = b^m \cdot (\prod p_i)^m$, ~~Consider~~

By choosing $k_i$ from the product of $m$ for prime $p_i$ in $(\prod p_i)^m$, writing the product is now $(\prod (p_i^{k_i})^{m/k_i})$, which is actually divisible by $n$, since $m/k_i \in \mathbb{Z}^+$, so the product becomes $a^m = b^m \cdot (n \cdot \prod (p_i^{k_i})^{m/k_i - 1})$, which is divisible by $n$, $\Rightarrow$ $m$ works, $a^m = 0$ ∎

c) Nilpotent elements of $\mathbb{Z}/72\mathbb{Z}$ are integers who have divisors of ~~2, 3~~ 2 and 3 $\Leftrightarrow$ the number is divisible by 6.

the set of nilpotent element $=$ ~~{6, 12~~ $\{0, 6, 12, \dots, 66\}$

1.c. Lemma: $a, b \in F$ where $F$ is a field. Then $ab = 0 \Rightarrow a = 0$ or $b = 0$

Pf: Either $a = 0$ or $a \neq 0$. If $a = 0$, we are done. If $a \neq 0$, $\exists a^{-1} \in F$. $\Rightarrow a^{-1}ab = a^{-1}0 \Rightarrow b = 0$ ∎

~~consider~~ Let $f \in R$, consider $f^m$, $x \in X$.

$f^m(x) = \underbrace{f(x) \cdot f(x) \dots f(x)}_{m \text{ times}}$. ~~By that~~ By applying the lemma above recursively, we conclude $f(x) = 0$ if $f^m(x) = 0$. $\forall x \in X$

This proves that if $f(x) \neq 0$, then $f(x)$ is not nilpotent.

2. a. $\exists k \in \mathbb{Z}^+$, $x^k = 0$. By well ordering principle, there exists a minimum $k'$ such that $x^{k'} = 0$. Now, either $x = 0$ or $x \neq 0$. If $x = 0$, we are done, if $x \neq 0$, then.

$x^{k'} = x \cdot x^{k'-1}$ where $k' - 1 \in \mathbb{Z}_{\geq 0}$ since $k' \in \mathbb{Z}_{>0}$.

BUT, $k' - 1 \neq 0$ otherwise $x = 0$ (contradiction) $\Rightarrow k' - 1 \in \mathbb{Z}_{>0}$.

By definition of $k'$, $x^{k'-1} \neq 0$. (for all $k < k'$, $x^k \neq 0$), which means $x \cdot x^{k-1} = 0$ where $x^{k-1} \neq 0$, meaning $x$ is a zero-divisor.

b. ~~Let~~ $\exists k \in \mathbb{Z}^+$, $x^k = 0$, Consider $(rx)^k$

$(rx)^k = \underbrace{rxrx \dots rx}_{k \text{ times}}$. Since $R$ is commutative, we can rearrange.

$(rx)^k = r^k x^k = r^k \cdot 0 = 0$ ∎

c. ~~Let~~ $\exists k \in \mathbb{Z}^+$, $x^k = 0$. Consider,

$(1+x)(1 - x + x^2 - x^3 + \dots (-1)^{k-1} x^{k-1})$, expanding, the middle terms cancel, leaving the expression equal to $1 + (-1)^{k-1} x^k$.

But $x^k = 0$, so that expression equals 1.

$\Rightarrow (1+x)$ is a unit.

d. Let $x$ be nilpotent and $y$ be a unit. Consider $(y^{-1}x + 1)$,

By part b, $y^{-1}x$ is nilpotent. By part c, $(y^{-1}x+1)$ is a unit, $\exists m : m(y^{-1}x+1) =$

Now, $my^{-1}y(y^{-1}x+1) = my^{-1}(x+y) = 1 \Rightarrow x+y$ is a unit

3. Let $I = \{(r,r) \mid r \in R\}$. $I$ is not empty ($R$ is assumed nonempty)

$(r_1, r_1), (r_2, r_2) \in \cancel{R} \; I$

① Closed under subtraction.

$$(r_1, r_1) - (r_2, r_2) = (r_1 - r_2, r_1 - r_2) \in I$$

② Closed under multiplication.

$$(r_1, r_1) \cdot (r_2, r_2) = (r_1 r_2, r_1 r_2) \in I \Rightarrow I \text{ is subring.}$$

Note: $r_1 - r_2 \in R$ and $r_1 r_2 \in R$ because $R$ is ring.

4. Show $O_f$ is subring.

① Closure: addition. $(a_1 + b_1 f w) + (a_2 + b_2 f w) = (a_1 + a_2) + (b_1 + b_2) f w \in O_f$

additive inverse: $-(a_1 + b_1 f w) = -a_1 - b_1 f w$ ✓

Closure: multiplication:

$$(a_1 + b_1 f w) \cdot (a_2 + b_2 f w) = a_1 a_2 + b_1 b_2 f^2 w^2 + f w (a_1 b_2 + a_2 b_1)$$

but $w^2 = \begin{cases} D & \text{if } D \equiv 2, 3 \bmod 4 \\ \left(\frac{1+\sqrt{D}}{2}\right)^2 & \text{if } D \equiv 1 \bmod 4 \end{cases}$

Consider $\left(\frac{1+\sqrt{D}}{2}\right)^2 = \frac{1+D+2\sqrt{D}}{4} = \frac{2+(D-1)+2\sqrt{D}}{4}$, and note $D \equiv 1 \pmod 4$,

so letting $k \in \mathbb{Z}$, $k = \frac{D-1}{4}$, $w^2 = k + w$.

$\Rightarrow$ $b_1 b_2 f^2 w^2$ will give either ~~or~~ the correct term to be absorbed into $fw$ coefficient or no coefficient.

Identity: $1 = a + bw$ where $a = 1$, and $b = 0$. $\Rightarrow$ exists!

Show $[O : O_f] = f$. By just focusing on the additive property, we notice every element in $O_f$ is uniquely determined by an ordered pair of 2 numbers, $a$ and $b$. $\Rightarrow (a, b)$.

With respect to the entire group $O$, the ~~second~~ coefficient to $w$ is $bf$, which is essentially, the ~~set~~ group $f\mathbb{Z}$.

Basically, $O = \mathbb{Z} \oplus \mathbb{Z}$, and $O_f = \mathbb{Z} \oplus f\mathbb{Z}$.

It's now easy to note the cosets being categorized by $\mathbb{Z}_f$, since ~~the cosets of~~ there are $f$ cosets of $f\mathbb{Z}$.

(The cosets are $0 + f\mathbb{Z}, 1 + f\mathbb{Z}, \ldots, (f-1) + f\mathbb{Z}$)

To prove the converse, note that you ~~are stuck with having a~~ can't have ~~anything~~ the first ~~term~~ constant term a to be anything but the integers (you can't have $a \in 2\mathbb{Z}$, ~~otherwise~~ because multiplication would not be closed, ~~thats~~ going back to $b_1 b_2 f^2 \omega^2$ term). ~~For~~ That means, for the index to be f, the final term $bf\omega$ has to have index f in $\mathbb{Z}$ (the one in $\mathcal{O}$) $\Rightarrow$ the ~~element~~ subgroup must be ~~$bf\mathbb{Z}$~~ $f\mathbb{Z}$ $\Rightarrow$ the form of (b ~~fw~~

5. a) $(-3r^2+rs)\times(r+r^2-2s) = \cancel{-3r^3} -3r^2\times(r+r^2-2s)+rs\times(r+r^2-2s)$

$= -3r^3-3r^4+6r^2s+rsr+rsr^2-2rs^2$

$= -3r^3-3+6r^2s+\cancel{rsr}+s+r^3s-2r$

b) $(r+r^2-2s)\times(r+r^2-2s) = r\times(r+r^2-2s)+r^2\times(r+r^2-2s) -2s\times(r+r^2-2s)$

$= r^2+r^3-2rs+r^3+r^4-2r^2s-2sr-2sr^2+4s^2$

$= r^2+2r^3-2rs+1-2r^2s-2r^3s-2r^2s+4$

$= 5+r^2+2r^3-2rs-4r^2s-2r^3s$

c) $\alpha\beta = -3-5r^3+7r^2s+r^3s$

$\alpha\beta-\beta\alpha = -2r^3+r^2s+2r-s$

d) $(\beta\alpha)\beta = (-3-2r+s-3r^3+6r^2s+r^3s)(-3r^2+rs)$

$= 9r^2+6r^3-3sr^2+9r-18r^2sr^2-3r^3sr^2$

$-3rs-2r^2s+srs-3r^4s+6r^2srs+r^3srs$

$= 9r^2+6r^3-3r^2s+9r-18s-3rs$

$-3rs-2r^2s+r^3-3s+6r+r^2$

$= -21s+15r+10r^2+7r^3-6rs-5r^2s.$

6 a) $\alpha+\beta = 6(1)+3(1\ 2)-3(2\ 3)+14(1\ 2\ 3)-7(1\ 3\ 2)$

b) $2\alpha-3\beta = -18(1)+6(1\ 2)-16(2\ 3)+28(1\ 2\ 3)+21(1\ 3\ 2)$

c) $(1\ 2)\cdot(2\ 3)=(1\ 2\ 3)$

$(1\ 2)(1\ 3\ 2)=(1\ 3)$

$(2\ 3)(2\ 3)=(1)$

$(2\ 3)(1\ 3\ 2)=(1\ 2)$

$(1\ 2\ 3)(2\ 3)=(1\ 2)$

$(1\ 2\ 3)(1\ 3\ 2)=(1)$

$\alpha\beta = 18(1\ 2)+6(1\ 2\ 3)-21(1\ 3)-30(2\ 3)-10(1)+35(1\ 2)+84(1\ 2\ 3)+28(1\ 2)-98(1)$
$= -108(1)+81(1\ 2)-21(1\ 3)-30(2\ 3)+90(1\ 2\ 3)$

d) $(2\ 3)(1\ 2)=(1\ 3\ 2)$

$(2\ 3)(1\ 2\ 3)=(1\ 3)$

$(1\ 3\ 2)(1\ 2)=(2\ 3)$

$(1\ 3\ 2)(2\ 3)=(1\ 3)$

$(1\ 3\ 2)(1\ 2\ 3)=(1)$

$\beta\alpha = 18(1\ 2)-30(2\ 3)+84(1\ 2\ 3)+6(1\ 3\ 2)-10(1)+28(1\ 3)-21(2\ 3)+35(1\ 3)-98(1)$
$= -108(1)+18(1\ 2)+63(1\ 3)-51(2\ 3)+84(1\ 2\ 3)+6(1\ 3\ 2)$

e) $(1\ 2)(1\ 2\ 3)=(2\ 3)$

$(1\ 2\ 3)(1\ 2)=(1\ 3)$

$\alpha^2 = (9+25+196)(1)-15(1\ 2\ 3)+42(2\ 3)-15(1\ 3\ 2)-70(1\ 3)+42(1\ 3)-70(1\ 2)$
$= 230(1)-70(1\ 2)-28(1\ 3)+42(2\ 3)-15(1\ 2\ 3)-15(1\ 3\ 2)$

7. Suppose $\varphi: 2\mathbb{Z}\to 3\mathbb{Z}$ was an isomorphism.
Consider $\varphi(4)=\varphi(2\times 2)=\varphi(2+2)$, let $3k=\varphi(2)$ for some $k\in\mathbb{Z}$

$\Rightarrow \varphi(2\times 2)=\varphi(2)\varphi(2)=9k^2$ since $\varphi(2)\in 3\mathbb{Z}$

$\varphi(2+2)=\varphi(2)+\varphi(2)=6k$

Now, $9k^2=6k$, the only solution of which is $k=0$.
But this implies ~~$\varphi(4)$~~ $\varphi(2)=0$, but since $\varphi$ is an isomorphism, it is injective, but ~~$\varphi(0)=0$~~ $\varphi(4)=0$ as well which is a contradiction

8 a). Does not satisfy multiplication. $A, B \in M_2(\mathbb{Z})$

$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ $B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$ $AB = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} & - \\ - & - \end{pmatrix}$

$\varphi(AB) = a_{11}b_{11} + a_{12}b_{21} \neq \varphi(A)\varphi(B) = a_{11}b_{11}$

b) $\varphi(AB) = a_{11}b_{11} + a_{12}b_{21} + a_{21}b_{12} + a_{22}b_{22}$ => Not equal

$\varphi(A)\varphi(B) = (a_{11} + a_{22})(b_{11} + b_{22})$

Not ~~homomorphism~~ Not homomorphism.

c). ~~$\varphi(A+B)$~~ = Show by counterexample:

$A = \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}$ $B = \begin{pmatrix} 3 & 2 \\ 4 & 1 \end{pmatrix}$ $\varphi(A) = 0$ $\varphi(B) = -5$

$\varphi(A+B) = \varphi\begin{pmatrix} 4 & 4 \\ 5 & 3 \end{pmatrix} = -8 \neq \varphi(A) + \varphi(B) = -5$

9) ~~Multiplication is not closed~~ For the following, denote $S$ as the set in question

a) $p, q \in S$ $p(0), q(0)$ are multiples of 3. $r \in \mathbb{Z}[x]$

Adding ~~the~~ two multiples of 3 is a multiple of 3.

=> closed by addition. Multiplying $p \cdot q$ has the effect of just multiplying an integer by a multiples of 3 for the constant term

=> closed by multiplication. $\boxed{\text{Ideal}}$. ~~Also any integer~~

b) $p(x) = 3x^2 + 2x$, $q(x) = 6x^2 + 2x$ $p, q \in S$

~~$pq =$~~ $x^2$ coefficient is 4 which is not multiple of 3. $\boxed{\text{Not Ideal}}$

c) This is pretty clearly an ideal.

d) Multiply any ~~element~~ polynomial in $\mathbb{Z}[x^2]$ by $x$, this is not ideal

e) ~~$p(x) = 2x^2 - x \in S$, $q(x) = 2x + 3 \in \mathbb{Z}[x]$~~

~~$pq = 2x^3 + x^2$~~ $4x^3 + 4x$ ~~katp~~

$S$ is clearly closed by addition.

Let $p = a_0 + a_1x + a_2x^2 + \dots + a_nx^n \in S$, and ~~q be a~~ $q \in \mathbb{Z}[x]$

~~p~~ $q = \sum b_i x^i$. $pq = \sum p b_i x^i$. ~~For each the~~ $p b_i x^i$ is a polynomial in $S$. By closure, so is $\sum p b_i x^i$ => $\boxed{\text{Ideal}}$

f. $p(x)=2 \in S$ . $q(x)=x \in \mathbb{Z}[x]$

$p \cdot q = 2x \notin S$ , $\boxed{\text{not ideal}}$

10. a. ~~Let $a \in I \cap J$ and choose an arbitrary $r \in R$.~~

~~$a \in I$ and $a \in J$. $ar \in I$ and $ar \in$~~

Let $a, b \in I \cap J$. $a \in I, b \in I$ and $a \in J, b \in J$

$\Rightarrow ab \in I \cap J$ closed under multiplication.

Let $r \in R$, $ar \in I$ because $a \in I$ and

$ar \in J$ " $a \in J$ $\Rightarrow ar \in I \cap J$.

Similarly, the same is true for the left ideal.

$\Rightarrow I \cap J$ is an ideal.

b. Let $I$ be the intersection of all ideals, it is clearly closed by addition,

$x, y \in I$ , where $x+y \in I$ since $x, y$ belong to every set in the collection

Let $r \in R$,

clearly for every set $rx$ is in that set, so $rx \in I$,

$\Rightarrow$ closed by left multiplication. Same is said for right multiplication.

PS 2 - 100BH.

1. Let $A := \{g-1 \mid g \in G\}$, and $I$ be the augmentation ideal, where augmentation homomorphism defined as.
$\varphi: RG \to R$, $\sum a_i g_i \to \sum a_i$.
We want to show ~~RG~~ $I = (A)$
(1) Show $I \subseteq (A)$;
Pf: For any $x \in I$, $\varphi(x) = 0$, $x = \sum a_i g_i$
$\Rightarrow \sum a_i = 0$
Consider $\sum a_i(g_i - 1)$, this is clearly an element in $(A)$.
$\sum a_i(g_i - 1) = \sum a_i g_i - a_i = \sum a_i g_i - \sum a_i$,
but this is just equal to $x. \Rightarrow x \in (A)$.
(2) Show $(A) \subseteq I$.
Pf: Let $x \in (A)$, $x = \sum a_i(g_i - 1)$, take the homomorphism...
$\varphi(x) = \sum \varphi(a_i)\varphi(g_i - 1)$, but $\varphi(g_i - 1) = 0$ since.
$g_i - 1 = 1 \cdot g_i - 1e$ where $e$ is the group identity
$\Rightarrow \varphi(g_i - 1) = 1 - 1 = 0 \Rightarrow \varphi(x) = 0 \Rightarrow x \in I$. ∎
(1), (2) $\Rightarrow (A) = I$

Now for the second part of the problem, let $G = \langle \sigma \rangle$. Every element in the augmentation ideal $I$ is generated by $A := \{g-1 \mid g \in G\} \Rightarrow$ for $x \in I$, ~~x ...~~
$x = a_1(\sigma - 1) + a_2(\sigma^2 - 1) + \dots + a_{n-1}(\sigma^{n-1} - 1)$, but
$\sigma^n - 1 = (\sigma - 1)(\sigma^{n-1} + \sigma^{n-2} + \dots + \sigma + 1)$, which means
~~x~~ x ~~can be factored into $\sigma - 1$~~ has a factor of $\sigma - 1$
$\Rightarrow \sigma - 1$ generates $I$.

2. (1). R is a field, Lemma: If an ideal $I$ contains a unit, $u$ ~~u is a unit in an ideal I~~, $I = R$.
Pf. of lemma: $\exists v$, $vu = 1$; Let $r \in R$ $r = r \cdot 1 = rvu = (rv)u \in I$ ∎
The converse clearly holds, since $1 \in R \Rightarrow 1 \in I$.
Lemma: $R$ is a field $\Leftrightarrow$ the only ideals of $R$ are $0$ and $R$.

Pf: R is a field $\Rightarrow$ every nonzero element is ~~an ideal~~ a unit.
~~All the possible ideals are generated by a single element.~~
~~Any ideal~~ Either the ideal is zero or nonzero. If its nonzero, it contains a nonzero unit $\Rightarrow$ the ideal equals R by 1st lemma. Conversely, if $u \neq 0$, $(u) = R \Rightarrow 1 \in R \Rightarrow \exists v: uv = 1$

If R is a field, it's ideals are 0 or R. Since $0 \neq R$, by definition 0 is a maximal ideal. Conversely, if 0 ~~and R are the maximal~~ is a maximal ideal, then the only ideal containing 0 ~~and R is~~ is 0 and R. Since every ideal is an additive subgroup, every ideal must contain $0 \Rightarrow$ every ideal is either 0 or $R \Rightarrow R$ is a field by lemma.

3. If $R/M$ is a field, every nonzero element is a unit. Any nonzero ideal ~~contains a non~~ of $R/M$ contains a unit. By lemma from q2 (which does not assume R commutative), any nonzero ideal equals $R/M$. By 4th isomorphism theorem, ~~$\exists$ bijection~~ every ideal of $R/M$ is a bijection to an ideal in R that contains M. So the ideals $I/M$ of $R/M$ can only be 0 or $R/M$. $\Rightarrow I/M = \cancel{M}$ or $I/R = R/M$. In which these imply ~~$I = R$~~. ~~M~~ $I = M$ or R. $\Rightarrow$ there is no ideal lying between M and $R \Rightarrow M$ is maximal.

4. (1) Suppose $(a) = (b)$, since R is an integral domain, it is commutative, $(a) = Ra = (b) = Rb \Rightarrow \exists u, v$ such that $a = ub$ and $b = va$. $\Rightarrow a = u(va) \Rightarrow a = (uv)a$. Since R is integral domain, $uv = 1$ or $a = 0$. The latter statement is true for unit 1, and $uv = 1$ implies u is a unit.

(2) Conversely, suppose $a = ub$, u is a unit. Ra is the set of multiples of a, ~~but r~~ so for $ra \in Ra$, $ra = r(ub) \in Rb$

$\Rightarrow Ra \subseteq Rb$. Since $u$ is unit, $u^{-1}$ exists, so $b = u^{-1}a$, by the same logic, $\Rightarrow Rb \subseteq Ra$.
$\Rightarrow Ra = Rb = (a) = (b)$. ■

5. ~~Since 0 is~~ $0 \in P$, suppose $ab = 0$ s.t. $a, b \in R$.
We want to show ~~$a \neq 0$ and $b \neq 0$~~ $a = 0$ or $b = 0$.
Since $0 \in P$, $\Rightarrow a \in P$ or $b \in P$. It suffices to show true for $a \in P$ ($b \in P$ follows by symmetry). If $a \in P$, then $a = 0$ or $a \neq 0$.
If $a = 0$, we are done. Suppose $a \neq 0$, then since $P$ contains no nonzero zero divisors $\Rightarrow b = 0$ ■

6. a

$$\begin{array}{r l}
 & 7x^9 + 101x^5 + 1621x \\
x^4 - 16 \,\big)\!\! & 7x^{13} - 11x^9 + 5x^5 - 2x^3 + 3 \\
 & -112x^9 \\
\hline
 & 101x^9 \\
 & -1616x^5 \\
\hline
 & 1621x^5 \\
 & -16 \cdot 1621x
\end{array}$$

$\Rightarrow \equiv 16 \cdot 1621x$. ~~$25936x$~~

$-2x^3 + 3 = \boxed{-2x^3 + 25936x + 3}$

b) $x^4 - 16 = (x^2 - 4)(x^2 + 4) = (x-2)(x+2)(x^2+4)$
mod $x^4 - 16 \Rightarrow (x-2)(x+2)(x^2+4) \equiv 0$
$\Rightarrow \overline{x-2}, \overline{x+2}$ are zero divisors.

7. The homomorphism ~~between~~ $\varphi: D^{-1}R \to \mathrm{Frac}\{R\}$ defined as $rd^{-1} \to rd^{-1}$ is an isomorphism between $D^{-1}R$ and ~~$\mathrm{im}\, D^{-1}R$~~ $\mathrm{im}\,\varphi$ where $\mathrm{im}\,\varphi$ is actually the subring. Lets show homomorphism.

(1) $\varphi(r_1d_1^{-1} + r_2d_2^{-1}) = \varphi\left(\dfrac{r_1d_2 + r_2d_1}{d_1d_2}\right) = \dfrac{r_1d_2 + r_2d_1}{d_1d_2} = r_1d_1^{-1} + r_2d_2^{-1}$

(2) $\varphi(r_1d_1^{-1} \cdot r_2d_2^{-1}) = \varphi\left(\dfrac{r_1r_2}{d_1d_2}\right) = \dfrac{r_1r_2}{d_1d_2} = r_1d_1^{-1}r_2d_2^{-1}$ ✓

im $\varphi$ is a subring, im $\varphi = \{ rd^{-1} \mid r \in R;\ d \in D \}$

(1) addition: let $r_1 d_1^{-1}, r_2 d_2^{-1} \in \mathrm{Im}\,\varphi$

$$r_1 d_1^{-1} + r_2 d^{-1} = \frac{r_1 d_2 + r_2 d_1}{d_1 d_2} \in \mathrm{Im}\,\varphi \quad \text{, because } d_1 d_2 \in D.$$

(2) Mult: $r_1 d_1^{-1} \cdot r_2 d_2^{-1} = \dfrac{r_1 r_2}{d_1 d_2} \in \mathrm{Im}\,\varphi$, becuse $d_1 d_2 \in D$. ✓

Now let's prove injectivity between im $\varphi$ and $D^{-1}R$.

Suppose $\varphi(\frac{r}{d}) = 0$, then $\frac{r}{d} = 0 \Leftrightarrow r = 0$

$\Rightarrow \dfrac{r}{d} = \dfrac{0}{d} \Rightarrow \ker \varphi = \{ \frac{0}{d} \}$

$\varphi$ is an isomorphism!

8. We can apply CRT to each coefficient separately.

Each polynomial ~~$f_i(x)$~~ has form.

$$f_i(x) = a_{i,0} + a_{i,1} x + \dots + a_{i,d} x^d.$$

Consider the $j^{th}$ coefficient of each of the $k$ polynomials,

~~The solution to~~ ~~$a_j$~~ By CRT, there exists a solution $a_j$ for

$$a_j \equiv a_{1,j} \pmod{n_1}$$
$$a_j \equiv a_{2,j} \pmod{n_2}$$
$$\vdots$$
$$a_j \equiv x_{k,j} \pmod{n_k}.$$

for every $j$, The polynomial $a_0 + a_1 x + a_2 x^2 + \dots + a_d x^d$ where coefficients exists by CRT, satisfy the system of modulos in the problem.

If every polynomial is monic, ie. $a_{1,d} = a_{2,d} = \dots = 1$, by CRT, $a_d = 1$ satifies the system of congruences

$\Rightarrow f(x)$ can be chosen to be monic.

9. a. $20 = 1\cdot 13 + 7 \quad d = 1 \quad \Rightarrow 20 - (13) + (7 - d)$

$13 = 1\cdot 7 + 6 \Rightarrow 13 - (7 - d) \quad 13 - (7 - d) = 7$

$7 = 1\cdot 6 + 1 \Rightarrow 7 - d = 1\cdot 6$

$6 = 6\cdot 1$

$20 = 1\cdot 13 + 7 \Rightarrow 2\cdot 20 = 2\cdot 13 + 13 + 1 \Rightarrow \boxed{2\cdot 20 - 3\cdot 13 = 1}$

$13 = 1\cdot 7 + 6 \Rightarrow 2\cdot 7 = 13 + 1$

$7 = 1\cdot 6 + 1 \Rightarrow 1\cdot 6 = 7 - 1$

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b. $372 = 5\cdot 69 + 27 \Rightarrow 5\cdot 372 = 25\cdot 69 + 2\cdot 69 - 3$

$69 = 2\cdot 27 + 15 \Rightarrow 2\cdot 69 = 4\cdot 27 + 27 + 3 \Rightarrow 2\cdot 69 - 3 = 5\cdot 27$

$27 = 1\cdot 15 + 12 = 2\cdot 15 - 3$

$15 = 1\cdot 12 + \underline{3} \Rightarrow 12 = 15 - 3$

$12 = 4\cdot 3$

$\Rightarrow \boxed{27\cdot 69 - 5\cdot 372 = 3}$

c. $\cancel{11391 = 1\cdot 5673 + 38\ 5718}$

$11391 = 2\cdot 5673 + 45 \Leftrightarrow 126\cdot 11391 = 252\cdot 5673 + 5673 - 3$

$5673 = 126\cdot 45 + 3 \Rightarrow 126\cdot 45 = 5673 - 3$

$45 = 15\cdot 3$

$\boxed{253\cdot 5673 - 126\cdot 11391 = 3}$

d. $507885 = 8\cdot 60808 + 21421 \Leftrightarrow \cancel{12\cdot 507885 = 96\cdot 60808 + 12\cdot 21421}$

$60808 = 2\cdot 21421 + 17966 \Leftrightarrow 6\cdot 60808 = 12\cdot 21421 + 5\cdot 21421 + 691$

$21421 = 1\cdot 17966 + 3455 \Leftrightarrow 5\cdot 21421 = 5\cdot 17966 + 5\cdot 3455$

$17966 = 5\cdot 3455 + 691 \Leftrightarrow 5\cdot 21421 = 6\cdot 17966 - 691$

$3455 = 5\cdot 691$

$17\cdot 507885 = 136\cdot 60808 + 6\cdot 60808 - 691$

$\Rightarrow \boxed{142\cdot 60808 - 17\cdot 507885 = 691}$

e). $\gcd = 1403109613$

$277522a - 32573b = d.$ ✓

10. a) $\gcd(2,4) = 2$ $\quad 2 \nmid 5 \Rightarrow$ no solution.

b). $\gcd(17,29) = 1$

$29 = 1\cdot 17 + 12 \Rightarrow 7\cdot 29 = 7\cdot 17 + 5\cdot 17 - 1$

$17 = 12\cdot 1 + 5 \Rightarrow 5\cdot 17 = 5\cdot 12 + 2\cdot 12 + 1 = 7\cdot 12 + 1$

$12 = 2\cdot 5 + 2 \Rightarrow 2\cdot 12 = 4\cdot 5 + 5 - 1 = 5\cdot 5 - 1$

$5 = 2\cdot 2 + \underline{1} \Rightarrow 2\cdot 2 = 5 - 1$

$2 = 2\cdot 1$

$12\cdot 17 - 7\cdot 29 = 1$

$17\cdot(31\cdot 12) + 29(-31\cdot 7) = 31$

$17(31\cdot 12 - x) = 29(31\cdot 7 + y)$

$17 \mid 31\cdot 7 + y$ and $29 \mid 31\cdot 12 - x$

$\Rightarrow 29n = 31\cdot 12 - x \Rightarrow \boxed{\begin{array}{l} x = 31\cdot 12 - 29n \\ y = 17n - 31\cdot 7. \end{array}}$

c). $145 = 1\cdot 85 + 60 \Rightarrow 7\cdot 145 = 7\cdot 85 + 5\cdot 85 - 5$

$85 = 1\cdot 60 + 25 \Rightarrow 5\cdot 85 = 5\cdot 60 + 2\cdot 60 + 5 = 7\cdot 60 + 5$

$60 = 2\cdot 25 + 10 \Rightarrow 2\cdot 60 = 4\cdot 25 + 25 - 5$

$25 = 2\cdot 10 + 5 \Rightarrow$

$10 = 2\cdot 5$

$12\cdot 85 - 7\cdot 145 = 5$

$85(12\cdot 101) + 145(-7\cdot 101) = 505$

$85(12\cdot 101 - x) = 145(7\cdot 101 + y)$

$17(12\cdot 101 - x) = 29(7\cdot 101 + y)$

$\boxed{\begin{array}{l} x = 12\cdot 101 - 29n \\ y = 17n - 7\cdot 101 \end{array}}$

PS 3.

1. Let R be ~~a~~ PID, and $I$ a prime ideal of R

Since $I$ is a prime ideal of a PID $\Rightarrow$ $I$ is maximal

Pf: Since R is PID, let $I=(p)$, we want to show the maximal ideal $J$ containing $(p)$ is either $(p)$ or $R$. Since $(p) \subseteq J$, $p=rm$ for some $r \in R$. Since $(p)$ is prime, either $r \in (p)$ or $m \in (p)$. If $m \in (p)$, then $(m) \subseteq (p) \Rightarrow (m)=(p)$. If $r \in (p)$, $r=ps$, $\Rightarrow p=psm \Rightarrow sm=1$ (R is integral domain) $\Rightarrow m$ is a unit. $\Rightarrow (m)=R$

Since $I$ is maximal, $R/I$ is a field as a corollary to Lattice Isomorphism Theorem. Since $R/I$ is field, any nonzero element is generated by $(1)$, because $a \neq 0$, $a \in R/I$ $\exists a^{-1}$: $1=a^{-1}a \Rightarrow a=a\cdot 1 \in (1)$. $0$ is generated by $(0)$, $R/I$ is a PID.

---

2. a) $I_2$: Suppose $I_2$ generated by $a_2+b_2\sqrt{-5}$.

$\Rightarrow \exists \alpha \in \mathbb{Z}[-5]$ : $2=\alpha(a_2+b_2\sqrt{-5})$ and

$\exists b \in \mathbb{Z}[-5]$ : $1+\sqrt{-5}=\beta(a_2+b_2\sqrt{-5})$. Define norm

$N(a+b\sqrt{-5})=a^2+5b^2$.

Lemma: $N([a+b\sqrt{-5}][c+d\sqrt{-5}])=N(a+b\sqrt{-5})N(c+d\sqrt{-5})$

$N(\cancel{ac-bd}\ ac-5bd+(ad+bc)\sqrt{-5})=(ac-5bd)^2+5(ad+bc)^2$

$=a^2c^2+25b^2d^2-\cancel{10abcd}+5a^2d^2+\cancel{10abcd}+5b^2c^2$

$N(a+b\sqrt{-5})N(c+d\sqrt{-5})=(a^2+5b^2)(c^2+5d^2)$

$=a^2c^2+5a^2d^2+5b^2c^2+25b^2d^2$ ✓

So, $N(2)=N(\alpha)N(a_2+b_2\sqrt{-5})=N(\alpha)(a_2^2+5b_2^2)=4$

$N(1+\sqrt{-5})=N(\beta)(a_2^2+5b_2^2)=6$

Possibilities: $N(\alpha)=+1 \Rightarrow \alpha=\pm 1 \Rightarrow \pm 2=a_2+b_2\sqrt{-5}$, which fails to satisfy $1+\sqrt{-5}=\pm 2\beta$ for any $\beta \in \mathbb{Z}[\sqrt{-5}]$

$N(\alpha) = 2 \Rightarrow a_2^2 + 5b_2^2 = 2$ (no integer solution)

$N(\alpha) = 4 \Rightarrow a_2^2 + 5b_2^2 = 1 \Rightarrow a_2 + b_2\sqrt{-5} = \pm 1$

$\Rightarrow a_2 + b_2\sqrt{-5}$ is a unit $\Rightarrow (a_2 + b_2\sqrt{-5}) = R$

$\exists \gamma, \delta \in R: \quad 2\gamma + \delta(1+\sqrt{-5}) = 1$

$(2-2\sqrt{-5})\gamma + \delta(+6) = 1 - \sqrt{-5}$

$\Rightarrow 2[(1-\sqrt{-5})\gamma + 3\delta] = 1 - \sqrt{-5}$

$\Rightarrow$ contradiction $\Rightarrow$ not principal idea

$I_3 \Rightarrow$ proved in textbook similarly.

$I_3' \Rightarrow \exists \alpha, \beta \in \mathbb{Z}[-5] : \alpha(a+b\sqrt{-5}) = 3 \quad , \beta(a+b\sqrt{-5}) = (2-\sqrt{-5})$

$\Rightarrow 9 = N(\alpha)(a^2+5b^2) \quad , \quad 9 = (a^2+5b^2)N(\beta)$

$N(\alpha) = 1 \Rightarrow \alpha = \pm 1 \Rightarrow a + b\sqrt{-5} = \pm 3$

$\Rightarrow \pm 3\beta = 2 - \sqrt{-5} \Rightarrow$ no integer solutions.

$N(\alpha) = 3 \Rightarrow a^2 + 5b^2 = 3 \Rightarrow$ no integer solutions $a, b$

$N(\alpha) = 9 \Rightarrow a^2 + 5b^2 = 1 \Rightarrow a + b\sqrt{-5} = \pm 1 \Rightarrow (a+b\sqrt{-5}) = \cancel{R}\ \mathbb{Z}[-5]$

$\exists \gamma, \delta \in \mathbb{Z}[-5]: \gamma 3 + \delta(2-\sqrt{-5}) = 1$

$\Rightarrow (6+3\sqrt{-5})\gamma + \delta(9) = 2 + \sqrt{-5}$

$\Rightarrow 3[\gamma(2+\sqrt{-5}) + 3\delta] = 2 + \sqrt{-5} \Rightarrow$ no solns.

b. ~~An~~ an element in $I_2^2$ is $[2a + b(1+\sqrt{-5})][2c + d(1+\sqrt{-5})]$

$= 4ac + bd(-4+2\sqrt{-5}) + 2ad(1+\sqrt{-5}) + 2bc(1+\sqrt{-5})$

$= 2(2ac - 2bd + ad + bc + \sqrt{-5}(bd + ad + bc)) \in (2)$

$\Rightarrow I_2^2 \subseteq (2)$

~~but~~ the generators of $I_2^2$ are $4, (1+\sqrt{-5})^2, 2(1+\sqrt{-5})$

$(1+\sqrt{-5})^2 = -4 + 2\sqrt{-5}$, $\Rightarrow -4 + 2\sqrt{-5} - 2 - 2\sqrt{-5} = -6$

so ~~every~~ ~~sa~~ considering generator of $-6$ and $4$,

~~we can se~~ $-6x + 4x = -2x$ for any $x \in \mathbb{Z}[-5]$

$\Rightarrow (2) \subseteq I_2^2 \Rightarrow \boxed{I_2^2 = (2)}$

c) $I_2 I_3$ generated by $(2a + (1+\sqrt{-5})b)(3c + d(2+\sqrt{-5}))$,

the generators are $(6,\ 4+2\sqrt{-5},\ 3+3\sqrt{-5},\ -3+3\sqrt{-5})$,

but since ~~$-3+3\sqrt{-5}$ is a generator $\Rightarrow I_2 I_3 \subseteq ($~~

Note $4+2\sqrt{-5} - (-3+3\sqrt{-5}) - 6 = 1-\sqrt{-5}$, so

we can achieve any $r\cdot(1-\sqrt{-5})$ with this combonation.

$\Rightarrow (1-\sqrt{-5}) \subseteq I_2 I_3$.

Now, note $6 = (1-\sqrt{-5})(1+\sqrt{-5})$

$4+2\sqrt{-5} = (1-\sqrt{-5})(-1+\sqrt{-5})$

$3+3\sqrt{-5} = (1-\sqrt{-5})(-2+\sqrt{-5})$

$-3+3\sqrt{-5} = -1(1-\sqrt{-5})$

$\Rightarrow$ every element in $I_2 I_3$ is a multiple of $(1-\sqrt{-5})$

$\Rightarrow I_2 I_3 \subseteq (1-\sqrt{-5}) \Rightarrow I_2 I_3 = (1-\sqrt{-5})$.

For $I_2 I_3'$... generators are $(6,\ 3+3\sqrt{-5},\ 4-2\sqrt{-5},\ 7+\sqrt{-5})$

Note: $(1+\sqrt{-5}) = (3+3\sqrt{-5}) + (4-2\sqrt{-5}) + (-6)$

$\Rightarrow (1+\sqrt{-5}) \subseteq I_2 I_3'$

Note: $6 = (1+\sqrt{-5})(1-\sqrt{-5})$ $\Rightarrow I_2 I_3' \subseteq (1+\sqrt{-5})$

$3+3\sqrt{-5} = (1+\sqrt{-5})\cdot 3$

$4-2\sqrt{-5} = (1+\sqrt{-5})(-1-\sqrt{-5})$

$7+\sqrt{-5} = (1+\sqrt{-5})(2-\sqrt{-5})$.

Since $R$ commutative, $I_2^2 I_3 I_3' = (I_2 I_3)(I_2 I_3')$

$\Leftrightarrow (1-\sqrt{-5})(1+\sqrt{-5}) = (6)$

3. a. By unique factorization, $\frac{p}{q} \in G$ has the form $2^e 3^f \frac{m}{n}$ where ~~$\frac{m}{n}$~~ ~~is~~ $m, n$ are relatively prime*. We let $e, f$ attain negative values to represent numerator and denominator.

Show $\varphi$ is a bijection.

~~$\varphi$~~ Surjective: $\varphi(g) = 2^e 3^f \frac{m}{n} \Rightarrow g = 2^f \cdot 3^e \cdot \frac{m}{n}$

$\Rightarrow \varphi$ is surjective.

* $m, n$ contain no factors of 2 or 3.

Injectivity: $\varphi(2^e \cdot 3^f \frac{m}{n}) = \varphi(2^{e'} \cdot 3^{f'} \cdot \frac{m'}{n'})$

$\Rightarrow 2^f \cdot 3^e \frac{m}{n} = 2^{f'} \cdot 3^{e'} \cdot \frac{m'}{n'}$

By uniqueness $\Rightarrow f = f'$, $e = e'$ $\Rightarrow \frac{m}{n} = \frac{m'}{n'} \Rightarrow m = m', n = n'$ (since $m, n$ relatively prime).

Show homomorphism:

$\varphi(\frac{p}{q} \cdot \frac{p'}{q'}) = \varphi(2^e \cdot 3^f \frac{m}{n} \cdot 2^{e'} 3^{f'} \cdot \frac{m'}{n'})$

$= 2^{f+f'} \cdot 3^{e+e'} \cdot \frac{m \cdot m'}{n \cdot n'}$

$\varphi(\frac{p}{q}) \cdot \varphi(\frac{p'}{q'}) = (2^f \cdot 3^e \cdot \frac{m}{n}) \cdot (2^{f'} \cdot 3^{e'} \cdot \frac{m'}{n'})$

$= 2^{f+f'} \cdot 3^{e+e'} \cdot \frac{m \cdot m'}{n \cdot n'} \Rightarrow \varphi$ is isomorphism

b. ~~Any map~~ Choose any two integer primes, and let $\varphi'$ be the map that functions the same as the function in part (a) by swapping the two chosen primes.

This map $\varphi'$ is an isomorphism as can be proved similarly above, substituting 2 and 3 for arbitrary prime $p$ and $q$.

Since there are infinitely many primes, there are infitely many isomorphisms.

c. Let $\varphi$: be the action of swapping primes $p_1$ and $p_2$.

To show $\varphi: \mathbb{Q} \to \mathbb{Q}$ is not a ring homorphism, let's compute some values, we expect $\varphi(\underbrace{p_1^e \cdot p_2^f \cdot \frac{m}{n} + p_1^e \cdot p_2^f \cdot \frac{m}{n} + \dots + p_1^e p_2^f \frac{m}{n}}_{p_1 \text{ times}}) =$

$= \underbrace{\varphi(p_1^e \cdot p_2^f \cdot \frac{m}{n}) + \varphi(p_1^e \cdot p_2^f \cdot \frac{m}{n}) + \dots + \varphi(p_1^e \cdot p_2^f \cdot \frac{m}{n})}_{p_1 \text{ times}}$

LHS: $\varphi(p_1 \cdot p_1^{e} \cdot p_2^{f} \frac{m}{n}) = \varphi(p_1^{e+1} p_2^{f} \cdot \frac{m}{n}) = p_1^{f} p_2^{e+1} \frac{m}{n}$

RHS: $\varphi(p_1^{e} \cdot p_2^{f} \cdot \frac{m}{n}) + \dots + \varphi(p_1^{e} p_2^{f} \cdot \frac{m}{n}) = p_1 \cdot p_1^{f} \cdot p_2^{e} \cdot \frac{m}{n}$

$= p_1^{f+1} p_2^{e} \cdot \frac{m}{n}$

LHS $\neq$ RHS, by uniqueness.

By definition, a ring homomorphism: $\varphi: \mathbb{Q} \to \mathbb{Q}$ must map $1_{\mathbb{Q}}$ to $1_{\mathbb{Q}}$

$\varphi(1) = 1$, by induction and the fact $\varphi(1+1) = \varphi(1) + \varphi(1)$

$\Rightarrow \varphi(n) = n$ where $n$ is a natural number, this is easily extended to 0 and the negative integers.

Now consider $\frac{a}{b} \in \mathbb{Q}$, $\varphi(\frac{a}{b}) = \varphi(a \cdot \frac{1}{b}) = \varphi(a) \cdot \varphi(\frac{1}{b}) = a\varphi(\frac{1}{b})$,

letting $a = b \Rightarrow \varphi(1) = b \cdot \varphi(\frac{1}{b}) \Rightarrow \varphi(\frac{1}{b}) = \frac{1}{b}$

Now in general, $\varphi(\frac{a}{b}) = \varphi(a) \cdot \varphi(\frac{1}{b}) = a \cdot \frac{1}{b} = \frac{a}{b} \Rightarrow$ identity.

4. $17 = 4^2 + 1^2$ since $17, 73, 101 \equiv 1 \mod 4$

$73 = 8^2 + 3^2$

$101 = 10^2 + 1$

$17 = (4+i)(4-i)$ ~~$(4+i)(8+3i) = 29 + 20i$~~ $\xrightarrow{10+i}$ ~~$270 + 229i$~~, $\xrightarrow{10-i}$ ~~$310 + 171i$~~

$73 = (8+3i)(8-3i)$ $(8+3i)(10+i)$

$101 = (10+i)(10-i)$ $= 77 + 38i$, $(8+3i)(10-i) = 83 + 22i$

$\Rightarrow n = (\pm 17 \cdot 77)^2 + (\pm 17 \cdot 38)^2 = (\pm 17 \cdot 83)^2 + (\pm 22 \cdot 17)^2$

Suppose now $(4+i)^2$ and $(4-i)^2$ ~~represent~~ divide the two squares,

$(4+i)^2(8 \pm 3i)(10 \pm i)$ are the remaining 4 possibilities

$(4+i)^2(8+3i)(10+i) = (15+8i)(77+38i) = (15 \cdot 77 - 8 \cdot 38) + i(8 \cdot 77 + 15 \cdot 38)$

$(4+i)^2(8+3i)(10-i) = (15+8i)(83+22i) = (15 \cdot 83 - 8 \cdot 22) + i(8 \cdot 83 + 15 \cdot 22)$

$(4+i)^2(8-3i)(10+i) = (15+8i)(83-22i) = (15 \cdot 83 + 8 \cdot 22) + i(8 \cdot 83 - 15 \cdot 22)$

$(4+i)^2(8-3i)(10-i) = (15+8i)(77-38i) = \underbrace{(15 \cdot 77 + 8 \cdot 38)}_{a} + i\underbrace{(8 \cdot 77 - 15 \cdot 38)}_{b}$

$\Rightarrow n = a^2 + b^2 \Rightarrow$ 6 distinct representations

& arrangements

$\shortparallel$

48 total representations

5. $n = \left(\frac{p}{q}\right)^2 + \left(\frac{p'}{q'}\right)^2 \Rightarrow \frac{p^2 + q^2}{q^2(q')^2} \Leftrightarrow n \cdot q^2 (q')^2 = p^2 + q^2$

~~$\Rightarrow$ the integer $n \cdot q^2 \cdot (q')^2 \equiv 1 \pmod 4$, the square of prime number $\equiv 1 \pmod 4$~~

~~$\Rightarrow n \equiv 1 \pmod 4$~~

~~Since~~ $n \cdot q^2 (q')^2$ is a sum of two squared integers, and ~~since~~ if $q, q' \equiv 3 \pmod 4$, $q, q'$ have an even (because they are squared) power $\Rightarrow$ if $q, q'$ divide $n$, ~~there~~ $n$ must have even powers of $q, q'$. $\Rightarrow n$ can be written as a sum of two squares.

6. a. Show $1+i$ is irreducible. $N(1+i) = 2$, if $1+i$ was reducible $1+i = \pi \cdot \pi' \Rightarrow N(\pi)N(\pi') = 2 \Rightarrow$ either $\pi$ or $\pi'$ is a unit, $\Rightarrow$ contradiction $\Rightarrow 1+i$ is irreducible.

Since $\mathbb{Z}[i]$ is Euclidean domain, ~~$(1+i)$~~ $1+i$ is prime, ~~and~~ $\Rightarrow (1+i)$ is maximal $\Rightarrow \mathbb{Z}[i]/(1+i)$ is a field.

~~Note:~~ ~~For~~ Quotienting out $(1+i)$, use division algorithm on arbitrary $\alpha \in \mathbb{Z}[i]$ $\alpha = q(1+i) + r$, $N(r) < N(1+i) = 2$

$\Rightarrow r = 0, 1, i \Rightarrow$ cosets are $0 + (1+i)$, $1 + (1+i)$, $i + (1+i)$.

But $1 + i(1+i) = i \in i + (1+i) \Rightarrow [1] = [i] \Rightarrow 2$ elements

$\Rightarrow \mathbb{Z}[i]/(1+i)$ has order 2.

b. Since $q \equiv 3 \pmod 4$, $q$ is irreducible. Following chain of logic above, $\mathbb{Z}[i]/(q)$ is a field. Consider the group structure of $\mathbb{Z}[i]/(q)$, $a + bi \equiv (a \bmod q) + (b \bmod q) i$

$\Rightarrow \mathbb{Z}[i]/q \cong \mathbb{Z}/q\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z}$ which has order $q^2$.

c. We need to show $(\pi)$ and $(\bar{\pi})$ are comaximal: $(\pi) + (\bar{\pi}) = \mathbb{Z}[i]$

We know $\pi$ is irreducible $\Rightarrow (\pi)$ is maximal since $\mathbb{Z}[i]$ is Euclidean domain.

Also, $(\pi)$ and $(\pi')$ are distinct ~~(don't d~~ (aren't associates)

$\Rightarrow (\pi) \subset (\pi) + (\bar{\pi}) = \mathbb{Z}[i]$ because $(\bar{\pi})$ is maximal

$\Rightarrow$ we can apply CRT $\Rightarrow \mathbb{Z}[i]/(\pi)(\bar{\pi}) = \mathbb{Z}[i]/(\pi\bar{\pi}) = \mathbb{Z}[i]/(p)$

on addition $\cong \mathbb{Z}[i]/(\pi) \times \mathbb{Z}[i]/(\bar{\pi})$

By group structure, $\mathbb{Z}[i] \cong \mathbb{Z} \times \mathbb{Z}$, and quotienting out by $(p)$

$\Rightarrow \mathbb{Z}[i]/(p) \cong \cancel{\mathbb{Z}[i]}\ \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z} \Rightarrow p^2 = |\mathbb{Z}[i]/(p)|$

Now, $|\mathbb{Z}[i]/(\pi)| = 1, p, p^2$, if $|\mathbb{Z}[i]/(\pi)| = 1$

~~$\Rightarrow (\bar{\pi}) = (p) \Rightarrow$ contradiction. $\bar{\pi}$ is a unit $\Rightarrow$ contradiction.~~

$\Rightarrow (\pi) = \mathbb{Z}[i] \Rightarrow \pi$ is a unit $\Rightarrow$ contradiction. Similarly,

if $|\mathbb{Z}[i]/(\pi)| = p^2 \Rightarrow \bar{\pi}$ is a unit $\Rightarrow$ contradiction.

$\Rightarrow |\mathbb{Z}[i]/(\pi)| = |\mathbb{Z}[i]/(\bar{\pi})| = p$, but are fields because $\pi$ and $\bar{\pi}$ are irreducible

7. a) $p(x) = 2y \cdot x^2 - 3y^3z \cdot x + 4y^2z^5$

$q(x) = \cancel{7z}\ \cancel{7+5y^3z^i}\ \cancel{(y^3}\ (5y^3z^4 - 3z^3 + 7)x^2$

b). $\deg(p) = 7$

$\deg(q) = 9$

c) $\deg_x(p) = 2 \quad \deg_x(q) = 2$

$\deg_y(p) = 3 \quad \deg_y(q) = 3$

$\deg_z(p) = 5 \quad \deg_z(q) = 4$

d) $pq = 14x^4y + 10x^4y^4z^4 - 6x^4yz^3 - 21x^3y^3z - 15x^3y^6z^5 + 9x^3y^3z^4$
$+ 28x^2y^2z^5 + 20x^2y^5z^9 - 12x^2y^2z^8$

$\deg_x(pq) = 4 \quad \deg_y(pq) = 6 \quad \deg_z(pq) = 9$

e). $pq = 20x^2y^5z^9 - 12x^2y^2z^8 + (-15x^3y^6 + 28x^2y^2)z^5 + (10x^4y^4 + 9x^3y^3)z^4$
$- 6x^4yz^3 - 21x^3y^3z + 14x^4y.$

8. Suppose $(x,y) = (f(x,y)) \Rightarrow \exists g(x,y) : f \cdot g = x$ by definition. ~~of ideal.~~

~~Taking~~ Since the $\deg_y(x) = 0$, $\Rightarrow$ the $\deg_y(f)$ must equal $0$ as well. Similarly, $\deg_x(f) = 0$. $\Rightarrow f(x,y) = c$, constant.

Since $c \in \mathbb{Q}$, $(c) = \mathbb{Q}[x,y]$, but ~~$\mathbb{Q}[x,y]$~~ $(x,y)$ doesn't generate constant terms.

9. a). $\mathbb{Z}[x]/(2) \cong \mathbb{Z}/2\mathbb{Z}[x]$, since you just mod all the coefficients, $\Rightarrow \mathbb{Z}[x]/(2)$ is Euclidean domain since $\mathbb{Z}/2\mathbb{Z}$ is a field. characteristic $= 2$

b). Equivalence classes in $\mathbb{Z}[x]/(x)$ ~~are the~~ are the constant term of the polynomial.

$p+(x) = q+(x) \Leftrightarrow p-q \in (x)$. $\Rightarrow \mathbb{Z}[x]/(x) \cong \mathbb{Z}$

since quotienting out by $(x)$ is equivalent to just the constant term.

c). $p+(x^2) = q+(x^2)$, ~~$\mathbb{Z}[x^2]/(x^2)$~~

$p = a_0 + a_1x + \dots + a_nx^n \quad q = b_0 + b_1x + \dots + b_mx^m$.

$p-q \in (x^2) \Rightarrow a_0 + a_1x - b_0 - b_1x + x^2(a_2 + \dots + a_nx^{n-2} - b_2 - \dots - b_mx^{m-2})$

$\Rightarrow$ equivalence classes are ~~isomorphic~~ ~~$\mathbb{Z}$~~ of form $a+bx$, only constant and linear term define class.

d) Cosets of $\mathbb{Z}[x,y]/(x^2,y^2,2)$ are of form ~~$fx^2+gy^2+2z$~~, where all coefficients are in $\mathbb{Z}/2\mathbb{Z}$.

$a + bx + cy + dxy$. Squaring a term like this,

~~means~~ $\Rightarrow (a+bx+cy+dxy)(a+bx+cy+dxy)$

$= a^2 + 2abx + 2acy + 2bcxy + O(x^2,y^2)$ where terms with $x^2$ or $y^2$ are equivalent to zero, since $2 \equiv 0$

$\Rightarrow$ the squared value $a^2 = a^2$ which is either 1 or 0

$a^2 = 0$ if $a \equiv 0$ (an even integer)

Characteristics: a) 2 b) 0 c) 0 d) 2

10. Suppose $R$ is a UFD, by Gauss' lemma, $\exists A, B \in F[x]$ such that $A(x) = ra$, $B(x) = sb$ for some $r,s \in F$

Since $p$ is monic, $sr = 1$, additionally, $rA \in R[x] \Rightarrow r \in R$ and $s \in R$ similarly. $\Rightarrow$ $r \in R^*$ $\Rightarrow a = r^{-1}A \in A$

$\mathbb{Q}[2\sqrt{2}] \ni x^2 - 2 = (x+\sqrt{2})(x-\sqrt{2})$ which is monic. $(x+\sqrt{2}) \in \mathbb{Q}[2\sqrt{2}]$

but $(x+\sqrt{2}) \notin \mathbb{Z}[2\sqrt{2}][x]$ since $\sqrt{2} \notin \mathbb{Z}[2\sqrt{2}]$

$\Rightarrow \mathbb{Z}[2\sqrt{2}]$ is not UFD.

PS 4.

1. ~~a~~ $f, g \in R$. $f-g \in R$ since $0x - 0x = 0x$
$fg \in R$, since the product of two constants have deg < 1.
and product of anything deg > [redacted] is > 1.
$\Rightarrow$ R is a subring.
$x^6 = x^2 \cdot x^2 \cdot x^2$, $x^2 \in R$.
$x^2$ is irreducible, ~~pf~~: suppose $fg = x^2$
$\deg f + \deg g = 2$, but $\deg f, \deg g \neq 1$,
$\Rightarrow$ one must be constant, the only constant such that
$a \cdot (bx^2) = x^2$ is if $a$ is a unit $\Rightarrow$ irreducible.
$x^6 = x^3 \cdot x^3$, similarly $x^3$ is irreducible, pf:
$fg = x^3 \Rightarrow$ $\deg f + \deg g = 3 \Rightarrow \deg f, \deg g \neq 1$
$\Rightarrow$ one must be a ~~unit~~ constant.
$\Rightarrow$ 2 different factorizations of irreducibles not associates.

2. a. Want to show ~~for any~~ $\forall x, y \in Tor(M), \forall r \in R$, $x + ry \in Tor(M)$.
~~Consider $x+ry$, multiply by $r_1 r_2$.~~
$\exists r_1, r_2 \in R$: $r_1 x = 0$, $r_2 y = 0$ by def. $r_1, r_2 \neq 0$ ~~$\Rightarrow r_1 r_2 (x+ry)$~~
Consider $r_1 r_2 (x + ry)$, $r_1 r_2 \neq 0$ since R is integral domain,
no zero-divisors. $r_1 r_2 (x+ry) = r_1 r_2 x + r_1 r_2 r y = 0 + 0 = 0$
$\Rightarrow x + ry \in Tor(M) \Rightarrow$ Submodule.

b. $\mathbb{Z}_6$. $Tor(\mathbb{Z}_6) = \{0, 2, 3, 4\}$, but $2+3 \notin Tor(\mathbb{Z}_6)$

c. If R has zero divisors, $\exists r_1, r_2$: $r_1 r_2 = 0$.
Consider an element ~~$m$, if $r_2 m = 0$, we are done.~~
$m \neq 0$. If $r_2 m = 0 \Rightarrow m \in Tor(M)$, we are done. Otherwise
$r_1(r_2 m) = 0 \Rightarrow r_2 m \in Tor$ and $r_2 m \neq 0$ ~~so $r_2 m$ is~~

3. Clearly closure by addition is satisfied $a_1, a_2 \in Ann(N) \Rightarrow a_1 + a_2 \in Ann(N)$.
① $Ann(N)$ is left ideal. $\forall r \in R$, $\forall n \in N$ ~~for~~ for ~~some~~ any $a \in Ann(N)$.
$ran = r0 = 0$ $\Rightarrow ra \in Ann(N)$
② $Ann(N)$ is right ideal. $a(rn) = 0$ $\Rightarrow Ann(N)$ is two sided
$rn \in N$

4. $Ann(I) := \{m \in M \mid am = 0 \quad \forall a \in I\}$.

Want to show ~~a~~ $x + ry \in Ann(I) \quad \forall x, y \in Ann(I), \forall r \in R$.

$\Leftrightarrow$ Showing $\forall a \in I$, $a(x+ry) = 0$, let $a \in I$

$$a(x+ry) = ax + ary = 0 + \underset{\substack{\Uparrow \\ I}}{(ar)}y = 0 \quad \blacksquare$$

$I$ since $a \in I$ and $I$ right ideal

5 a. We must find $lcm(24, 15, 50) = 3 \cdot 8 \cdot 25 = 600$

~~Ideal~~ $Ann(M) = (600)$.

b. $m = (m_1, m_2, m_3)$ , Similarly. $15\mathbb{Z}$ and $25\mathbb{Z}$.

$\Uparrow$ Since $2 \nmid 15$.

$am_1 \equiv 0 \pmod{24}$.

$\Rightarrow m_1 \equiv 12 \pmod{24}$.

~~agn~~ $\Rightarrow$ consider $12\mathbb{Z}$.

$$Ann(I) \cong 12\mathbb{Z} \times 15\mathbb{Z} \times 25\mathbb{Z}.$$
$$\cong \mathbb{Z}/2\mathbb{Z} \times \{0\} \times \mathbb{Z}/2\mathbb{Z}.$$

6 Since $M$ is finite, let $|M| = n$. We know for any $a \in M$, ~~$n \in \mathbb{Z}$~~ $na = 0$, By Lagrange's. ~~We expect~~

If extended to $\mathbb{Q}$-Module, let $a \neq 0$, $0 = n(\frac{1}{n}a) = a \Rightarrow a = 0$ for all $a$, which is the trivial module, contradicting that $a \neq 0$.

7. A homomorphism $\varphi \in Hom(\mathbb{Z}/n\mathbb{Z}; \mathbb{Z}/m\mathbb{Z})$ is determined by the element $\varphi(1)$ maps to, since we can obtain $\varphi(k)$ by additivity $\varphi(k) = k\varphi(1) = \varphi(1) + \ldots + \varphi(1)$.

Let $\varphi(1) = k$. We know valid homomorphism must preserve $\varphi(n) = \varphi(0) = 0 \Rightarrow \varphi(n) = n\varphi(1) = 0 \Rightarrow ord\, \varphi(1) \mid n$.

$\Leftrightarrow nk \equiv 0 \pmod{m} \Rightarrow \exists r: \quad nk = mr$. Let $d = \gcd(m, n)$.

$n' = \frac{n}{d}$, $m' = \frac{m}{d}$, $\gcd(n', m') = 1$. $\Rightarrow n'k = m'r \Rightarrow m' \mid k$

$\Rightarrow \underbrace{\frac{m}{\gcd(n,m)}}_{m'} \mid k \Rightarrow k = i\frac{m}{d}$ for $i \in \mathbb{Z}$. The possible values of $k$ form a cyclic subgroup of $\mathbb{Z}/m\mathbb{Z}$, each corresponding to a homomorphism.

$\Rightarrow$ the order of the subgroup is $d \Rightarrow Hom(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/m\mathbb{Z}) \cong \mathbb{Z}/d\mathbb{Z}$.

8. Let $\phi: R \to Hom_R(R,R)$ be mapping $r \to rI$ where $I(x) = x$ is the identity functions.

Show $\phi$ is a ring homomorphism.

① Addition: $\phi(r_1+r_2) = (r_1+r_2)I(x) = I((r_1+r_2)x)$
$= r_1x + r_2x = I(r_1x) + I(r_2x)$.
$= \phi(r_1) + \phi(r_2)$.

② Mult: ~~$\phi(r_1r_2)$~~ define multiplication in $Hom_R(R,R)$ as composition
$\phi(r_1r_2) = (r_1r_2)I(x) = r_1 I(r_2x)$
$= I(r_1 I(r_2x)) = \phi(r_1)\cdot\phi(r_2)$.
$= r_1 I(r_2 I(x))$

③ Id: $\phi(1) = 1I = I$ ✓

Show injective.

Suppose $\phi(r) = 0 \Rightarrow rI(x) = 0 \;\; \forall x \in R$.
$\Leftrightarrow rx = 0 \;\; \forall x \in R$, including $x = 1$.
$\Rightarrow r = 0 \Rightarrow \ker\phi = \{0\} \Rightarrow \phi$ is injective.

Surjective: ~~If~~ ~~rI...~~ This is super obvious $rI = \phi(r)$, $\forall r \in R$.

$\Rightarrow R \cong Hom_R(R,R)$. 9. Examples $\mathbb{Q}/\mathbb{Z}$.

9. Let ~~M~~ $M$ be finitely generated torsion R-module by ~~M~~ $m_1, m_2, \dots, m_n$. $M = Rm_1 + Rm_2 + \dots + Rm_n$.
Since $M$ is torsion R-module, $\exists r_1, \dots, r_n$: $r_i \neq 0$ $r_1m_1 = 0, r_2m_2 = 0, \dots, r_nm_n = 0$. Let $r = r_1r_2\dots r_n \neq 0$ since $R$ is int domain.
$\forall m \in M$: $m = a_1m_1 + \dots + a_nm_n$. $rm = 0$. ∎

10. Since $M/N$ is finitely generated, it's generators are $\overline{m_i} = m_i + N$.
~~Let $x \in M$~~ Let $N$ be generated by $\{n_1, \dots, n_n\}$.
$x + N = \overline{x} = \sum a_i\overline{m_i}$ ~~but $x \in \sum b_i n_i$~~ $\Rightarrow 1.x - \sum a_i\overline{m_i} \in N$
$\Rightarrow x - \sum a_i\overline{m_i} = \sum b_in_i \Rightarrow x = \sum a_im_i + \sum b_in_i \Rightarrow M$ generated by $\{n_1, \dots, n_n, m_1, \dots, m_m\}$

PS5

1. Suppose $M$ irreducible. Let ~~a≠0 be~~ $a \in M$ be $a \neq 0$.
Consider the submodule generated by $a$. We can check $Ra$ is a submodule easily.
$x, y \in Ra, r \in R$ $\quad x = r_1 a, y = r_2 a$, for some $r_1, r_2 \in R$
$\Rightarrow \quad r_1 a + r r_2 a = (r_1 + r r_2) a \in Ra$.
But the only submodules possible are $0$ or $M$,
$\Rightarrow Ra = 0, Ra = M$. ~~If~~ But $Ra = 0$ iff $a = 0$,
$\Rightarrow Ra = M$ for $\forall a \in M$.
Conversely, if $M = Ra$ $\forall a \in M$. Consider submodule $N \subseteq M$.
Let ~~n≠0~~ $n \in N$ be nonzero element in $N$. By assumption,
$Rn = M$, $\Rightarrow M = Rn \subseteq N$. $\Rightarrow M = N$. Every submodule is either $M$ or the $0$ submodule $\Rightarrow$ irreducible.
$\mathbb{Z}$ modules are abelian groups, so irreducible $\mathbb{Z}$-modules $\Leftrightarrow$ all g. subgroups are either $0$ or the group itself
$\Rightarrow$ group of prime order. (by Lagrange's, all subgroups must divide $p$).

2. ~~Let $\varphi: R \to M$ defined by $r \to rm$.~~
Let $\varphi_m: R \to M$ defined by $r \to rm$. From the previous problem, we know $\operatorname{im} \varphi_m = M$. By 1[st] isomorphism thm,
$\Rightarrow R/\ker \varphi_m \cong \operatorname{im} \varphi_m = M$. We ~~can~~ ~~obtain~~ show $\ker \varphi_m = I$ is maximal. The correspondence theorem states ∃bijection between ideals containing ~~ker φm~~ $I$ in $R$ and the ideals of $R/I$. Since $R/I \cong M$, $R/I$ must also ~~contain~~ have no proper submodules beside $0$ $\Rightarrow$ these two submodules must correspond to the ~~submodul~~ "ideals" ($R$-submodules) containing $I$ $\Rightarrow$ ~~I, R~~ they must be $I$ and $R$ itself $\Rightarrow I$ maximal.
Conversely, let $M \cong R/I$, where $I$ is maximal ideal in $R$. By correspondence thm, the same bijection ~~means since~~ implies the two ideals in $R$ containing $I$ corresponds with 2 ideals of $R/I$, $0$ and $R/I$ itself $\Rightarrow$ submodules of $M$ are $0$ and $M$ $\Rightarrow$ irred.

3. Let $\varphi: M_1 \to M_2$, be a nonzero ~~homo~~ R-module homomorphism.
Since $\ker \varphi$ is submodule of $M_1$, $\ker \varphi = 0$, or $M_1$.
$\ker \varphi \neq M_1$, otherwise $\varphi = 0$ is the zero homomorphism.
$\Rightarrow \ker \varphi = 0 \Rightarrow \varphi$ is injective.
~~im~~ By similar reasoning, $\operatorname{im} \varphi = M_2 \Rightarrow \varphi$ is surjective.
$\Rightarrow \varphi$ is isomorphism.
If $M$ is irred, then $\operatorname{End}_R(M)$ is a set of isomorphisms,
for every $\phi \in \operatorname{End}_R(M)$, $\exists \phi^{-1} \Rightarrow \operatorname{End}_R(M)$ is division ring.

4. Clearly $\forall m \in M$, $m = em + (1-e)m \Rightarrow M = eM + (1-e)M$
~~Let first~~ By showing $eM \cap (1-e)M = \{0\}$, we show $M = eM \oplus (1-e)M$
Suppose $x \in eM \cap (1-e)M$, $x = em_1$ and $x = (1-e)m_2$.
$ex = e(em) = em = x$, $ex = e(1-e)m_2 = (e-e^2)m_2 = 0m_2 = 0$.
$\Rightarrow x = 0$, $\Rightarrow M = eM \oplus (1-e)M$

5. Let $M_i$ be a free module, we want to show $\oplus M_i$ is free R-module as well. Let $B_i$ be the generating set for $M_i$,
I claim $\bigcup B_i = B$ ~~spa.~~ generates $\oplus M_i$ and is linearly independent.
For $x \in \oplus B_i$, $x = \sum x_j$, where $x_j$ can be written uniquely since $M_j$ is free. $\Rightarrow$ spanning.
Suppose $x = 0$, by the uniqueness of $B_i$, we require $x_i = 0$ as well $\Rightarrow$ linear independence.

6. $2 \otimes 1 = 2 \cdot 1 \otimes 1 = 1 \otimes 1 \cdot 2 = 1 \otimes 0 = 0$ in $\mathbb{Z} \otimes \mathbb{Z}/2\mathbb{Z}$.
$1 \otimes 0 = 0$ since the coset containing $(1,0)$ is the identity
~~based on~~ since $m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2$
In $2\mathbb{Z} \otimes \mathbb{Z}/2\mathbb{Z}$, $2$ is a basis element, we can't factor $2 = 2 \cdot 1$ since ~~1 is~~ $1 \notin 2\mathbb{Z}$.

7 $\mathbb{C} \otimes_R \mathbb{C}$ is left R-module by the action.

$r(m \otimes n) = (rm) \otimes n$

$\mathbb{C} \otimes_{\mathbb{C}} \mathbb{C}$ is left R-module by the same action and restrict $\mathbb{C}$ to the set R.

They are not isomorphic since we have two extra degrees of freedom for $\mathbb{C} \otimes_{\mathbb{C}} \mathbb{C}$, in allowing factorizations, such as $2 = (1+i)(1-i)$ as ~~potentia~~ means of "sliding" terms.

8. Let ~~$\bar{x} \otimes \bar{y} \in (\mathbb{Q}/R) \otimes_R (\mathbb{Q}/R)$.~~

~~$x = \frac{x}{d} + R \iff x + R = d(\frac{x}{d}) + R$~~

~~$= x \otimes y = [d(\frac{x}{d} + R)] \otimes [\frac{y}{d} + R]$~~

Consider $(x+R) \otimes_R (y+R) \in (\mathbb{Q}/R) \otimes_R (\mathbb{Q}/R)$

~~if~~ we can divide $x$ by $d$, let $y = \frac{c}{d}$ for some $c$.

$(d(\frac{x}{d}+R)) \otimes (\frac{c}{d}+R)$

$\Rightarrow (\frac{x}{d}+R) \otimes c+R$, but since $c \in R$, then $c+R=0$.

$= 0$.

9. ~~Prove that~~ Every $n \in N$ can be written as unique summation of basis, $n = \sum a_i e_i$.

So, an element $m \otimes n \in M \otimes N$ can be written as $m \otimes \sum a_i n_i$, which is expanded to $\sum (m \otimes a_i n_i) = \sum a_i m \otimes n_i = \sum m_i \otimes n_i$. since M is R-module as well. If the sum is 0, $m_i$ must be 0, since $n_i$ was assumed to be nonzero. Additionally, since, the uniqueness property of direct sums was propagated through the tensor product.

b. $1 \otimes 2 = 1 \otimes (2 \times 1) = 2 \otimes 1 = 0 \otimes 1 = 0$. in this case. $m=1, n=2$, but $n=2$ is linearly independent.

10. Suppose we could write it as $v \otimes w$

$v = ae_1 + be_2$, $w = ce_1 + de_2$, then

$$v \otimes w = (ae_1 + be_2) \otimes (ce_1 + de_2) = ac(e_1 \otimes e_1) + ad(e_1 \otimes e_2)$$
$$+ bc(e_2 \otimes e_1) + bd(e_2 \otimes e_2)$$

$\Rightarrow ac = 0, bd = 0 \quad ad = 1 \quad bc = 1.$

$ad = 1 \Rightarrow a \neq 0, d \neq 0$, $bc = 1 \Rightarrow b \neq 0, c \neq 0$,

then $ac = bd = 0 \Rightarrow$ contradiction

PS6

1. ($\Leftarrow$), if $v = av'$ for some $a \in F$,

~~we~~ $\cancel{v \otimes v' = v' \otimes v}$

$$v' \otimes v = v' \otimes av' = v'a \otimes v' = av' \otimes v' = v \otimes v'$$

($\Rightarrow$), if $v \otimes v' = v' \otimes v$, $v \otimes v' - v' \otimes v = 0$

$$\cancel{v \otimes v' - v' \otimes v = (v - v') \otimes (v' - v) = 0}$$

$$\Rightarrow \cancel{v = v'}$$

Construct a basis of $V$ by extending $\{v, v'\}$, (let's assume they are linear independent, and eventually show a contradiction). $B = \{v, v'\} \cup \{e_i\}_{i \in I}$. Now, the basis of $V \otimes V$ ~~is~~ is $\{b_i \otimes b_j \mid b_i, b_j \in B\}$, of which ~~we~~ $v \otimes v'$ and $v' \otimes v$ are elements of the basis, but since they are linearly independent,

$\Rightarrow v \otimes v' \neq v' \otimes v$, a contradiction, $\Rightarrow v, v'$ are linearly dependent.

2. Show $(\prod M_i) \otimes_{\mathbb{Z}} \mathbb{Q} \neq \prod (M_i \otimes_{\mathbb{Z}} \mathbb{Q})$

RHS: For $M_i \otimes_{\mathbb{Z}} \mathbb{Q}$, let ~~$m \otimes$~~ $m \otimes \frac{a}{b} \in M_i \otimes_{\mathbb{Z}} \mathbb{Q}$.

$$m \otimes \frac{a}{b} = m \otimes \frac{2^i a}{2^i b} = 2^i m \otimes \frac{a}{2^i b} = 0$$

$\Rightarrow$ RHS $= 0$.

LHS: Letting $x = (1, 1, \dots, ) \in \prod M_i$, we see

$x \otimes \frac{a}{b} \neq 0$, ~~otherwise, ∃ an integer that divides~~ and there is no integer such that $nx = 0$, $\Rightarrow$ can't be equal to RHS.

3. ~~If~~ Suppose $rm = 0$, with $0 \neq r \in R$ and $m \in I \otimes_R I$, let $I = (a)$, ~~$m = r_1 a \otimes r_2 a$. If rm~~

$$\cancel{0 = rm = r(r_1 a \otimes r_2 a) = r r_1 a \otimes r_2 a}$$

Define map $R \to I$, ~~$r \to ra$~~ $r \to ra$. This is a bijective homomorphism $\Rightarrow R \cong I$ as $R$-modules. $\Rightarrow I \otimes_R I \cong R \otimes_R R \cong R$

Since $R$ is integral domain, $ra = 0 \Rightarrow r = 0$ or $a = 0$,

$\Rightarrow$ no torsion elements.

4. $2(2\otimes x - x\otimes 2) = 4\otimes x - 2x\otimes 2$
$= 2\cdot 2\otimes x - 2x\otimes 2 = 2\otimes 2(x) - 2x\otimes 2$.
$= 2x\otimes 2 - 2x\otimes 2 = 0$.

$x(2\otimes x - x\otimes 2) = 2x\otimes x - x\cdot x\otimes 2 = 2x\otimes x - x\otimes 2x$
$= 2x\otimes x - 2x\otimes x = 0$

Define $\varphi: R \to I\otimes_R I$ as $\varphi(r) = r(2\otimes x - x\otimes 2)$.
Earlier, we found $\varphi(2) = \varphi(x) = 0$. $\Rightarrow (2,x) = I \subseteq \ker\varphi$.
But since $\mathbb{Z}[x]/I \cong \mathbb{Z}/2\mathbb{Z}$, $\ker\varphi$ is maximal, and $\ker\varphi \neq R$
$\Rightarrow \ker\varphi = I$. $\Rightarrow$ $R/I \cong$ ~~such~~ $\operatorname{im}\varphi$ = subgroup generated by $(2\otimes x - x\otimes 2)$.

5. Suppose $2\otimes 2 + x\otimes x = a\otimes b$ $\quad a,b\in I$.
Let $\varphi: I\times I \to R$ be such that $\varphi(a,b) = ab$.
this is ~~R-balanced~~ ~~induces~~ bilinear $\Rightarrow$ induces
R-module homomorphism $\phi: I\otimes I \to R$ $\phi(a\otimes b) = \varphi(a,b)$.
$\phi(2\otimes 2 + x\otimes x) = \phi(2\otimes 2) + \phi(x\otimes x) = 4 + x^2 = ab$.
$a = ux + 2v \qquad b = u'x + 2v'$
$ab = uu'x^2 + 4vv' + (2v'u + 2u'v)x = 4 + x^2$.
$\Rightarrow uu' = 1 \Rightarrow u = u' = \pm 1$. $\quad 4 + x^2 = x^2 + 4vv' + (2v' + 2v)x$.
$\Rightarrow \quad v' + v = 0, \quad vv' = 1 \Rightarrow$ contradiction.

6. a. Since $\varphi$ surjective, for $c \in \ker\gamma \subseteq C$, $\exists b: \varphi(b) = c$.
Since diagram commutes, going $B \to C \to C'$ is the same as $B \to B' \to C'$
$\Rightarrow \varphi'(\beta(b)) = 0$ since $\gamma(\varphi(b)) = 0$.
This implies $\beta(b) \in \ker\varphi' = \operatorname{im}\psi'$ since rows are exact.
$\Rightarrow \exists a' \in A'$: $\psi'(a') = \beta(b)$. Since $\alpha$ is surjective, $\exists a \in A \mid \alpha(a) = a'$.
Since diagram commutes we can obtain $\beta(b)$ by $A \to A' \to B'$ or $A \to B \to B'$, the latter being $\beta(\psi(a)) = \beta(b)$, since $\beta$ injective, $\psi(a) = b$. $\Rightarrow c = \varphi(\psi(a))$, But $\ker\varphi = \operatorname{im}\psi$. $\Rightarrow c = 0 \Rightarrow \gamma$ is injective

b. $b \in \ker \beta \subseteq B$. We know by commutativity

$\gamma(\varphi(b)) = \varphi'(\beta(b)) = \varphi'(0) = 0$

$\Rightarrow \varphi(b) \in \ker \gamma$, but $\gamma$ is injective $\Rightarrow \varphi(b) = 0$

$\Rightarrow b \in \ker \varphi = \operatorname{im} \psi \Rightarrow \exists a \in A \mid \psi(a) = b$.

Now $A \to B \to B'$ and $A \to A' \to B'$.

$\beta(\psi(a)) = \psi'(\alpha(a)) = \beta(b) = 0 \Rightarrow \alpha(a) \in \ker \psi'$

$\Rightarrow \alpha(a) = 0$ since $\psi'$ injective, but $\alpha$ is also injective

$\Rightarrow a = 0 \in \ker \alpha$. $\Rightarrow \psi(a) = 0 = b$.

c. Let $b' \in B'$, WTS $\exists b : \beta(b) = b'$.

$\varphi'(b') = c'$. $\exists c : \gamma(c) = c' = \varphi'(b')$.

$\exists b : \varphi(b) = c \Rightarrow \gamma(\varphi(b)) = \varphi'(b')$.

By commutativity, $\varphi'(\beta(b)) = \gamma(\varphi(b)) = \cancel{\varphi'(b')\ \gamma(c)}$

$= \varphi'(b')$.

Since $\varphi'(\beta(b) - b') = 0 \Rightarrow (\cancel{\psi}\,\beta(b) - b') \in \ker \varphi'$

$\Rightarrow \exists a' : \psi'(a') = \beta(b) - b'$, $\exists a : \alpha(a) = a'$

Commutativity: $\psi'(\alpha(a)) = \beta(\psi(a))$, Let $\psi(a) = b_1$

$\Rightarrow \beta(b) - b' = \beta(b_1) \Rightarrow b' = \beta(b) - \beta(b_1) = \beta(b - b_1)$.

$\Rightarrow$ We found $b - b_1$

d. $c \in \ker \gamma$, $\gamma(c) = 0 \Rightarrow$ $\cancel{\varphi'(\gamma(c)) = 0 \Rightarrow \gamma(c) \in \ker \varphi'}$

$\alpha$ and $\varphi$ are surjective. $\exists b \in B \mid \varphi(b) = c$.

~~Com.~~ $B \to C \to C' = B \to B' \to C' \Rightarrow \gamma(\varphi(b)) = \varphi'(\beta(b)) = 0$

$\Rightarrow \beta(b) \in \ker \varphi' \Rightarrow$ by exactness $\exists a' : \psi'(a') = \beta(b)$.

$\exists a : \alpha(a) = a'$ by surjectivity, $A \to B \to B' = A \to A' \to B'$

$\Rightarrow \beta(\psi(a)) = \psi'(\alpha(a)) = \beta(b) \Rightarrow \psi(a) = b$.

$\Rightarrow \varphi(\psi(a)) = c = 0$ since $\psi(a) \in \operatorname{im} \psi = \ker \varphi$.

e. Let $a' \in A'$, $\psi'(a') \in \operatorname{im}\psi' = \ker\varphi'$, let $\psi'(a') = b'$

$\varphi'(b') = 0$. Since $\beta$ surjective, $\exists b: \beta(b) = b'$

$B \to C \to C' = B \to B' \to C' \Rightarrow \gamma(\varphi(b)) = \varphi'(\beta(b)) = 0$.

~~$\Rightarrow \varphi(b) \in \ker\varphi = \operatorname{im}\psi \Rightarrow \exists a: \psi(a) = \varphi(b)$~~.

~~$A \to A' \to B' = A \to B \to B' \Rightarrow \beta(\psi(a)) = \psi'(\alpha(a))$~~

Since $\gamma$ injective $\varphi(b) = 0$. $\Rightarrow b \in \ker\varphi = \operatorname{im}\psi$.

$\Rightarrow \exists a: \psi(a) = b$.

$A \to A' \to B' = A \to B \to B' \Rightarrow \beta(\psi(a)) = \psi'(\alpha(a))$

$\beta(\psi(a)) = b' = \psi'(a') = \psi'(\alpha(a)) \Rightarrow \alpha(a) = a'$

$\Rightarrow$ we found a!

$$\begin{tikzcd} A \arrow[r, "x"] \arrow[d, dash] & B \arrow[r, "y"] \arrow[d, dash] & C \arrow[r, "z"] \arrow[d, dash] & D \arrow[d, dash] \\ A' \arrow[r, dash, "x'"'] & B' \arrow[r, dash, "y'"'] & C' \arrow[r, dash, "z'"'] & D' \end{tikzcd}$$

7\. a. $c \in \ker\gamma$.

$z(c) = d$, $\delta(d) = d'$

$\delta(z(c)) = z'(\gamma(c)) = z'(0) = 0$

$\Rightarrow z(c) = 0 \Rightarrow c \in \ker z \Rightarrow \exists b: y(b) = c$.

$y'(\beta(b)) = \gamma(y(b)) = 0. \Rightarrow \beta(b) = b' \in \ker y' \Rightarrow \exists a': x'(a') = b'$

$\exists a: \alpha(a) = a' \Rightarrow x'(\alpha(a)) = \beta(x(a)) = b' = \beta(b)$

~~$y'(\beta(b)) =$~~ $\Rightarrow x(a) = b$ since $\beta$ injective.

$c = y(b) = y(x(a)) = 0$ since $x(a) \in \operatorname{im} x = \ker y$

b. Let $b' \in B'$, $c' = y'(b')$, $\exists c: \gamma(c) = c' = y'(b')$.

$z'(y'(b')) = 0$ since $y'(b') \in \operatorname{im} y' = \ker z'$

$C, D, D' = C, C', D' \Rightarrow \delta(z(c)) = z'(\gamma(c)) = 0. \Rightarrow z(c) = 0$ since

$\delta$ injective.. $\Rightarrow c \in \ker z = \operatorname{im} y \Rightarrow \exists b: y(b) = c$.

$B, C, C' = B, B', C' \Rightarrow \gamma(y(b)) = y'(\beta(b)) = y'(b')$

$\Rightarrow \beta(b) - b' \in \ker y' = \operatorname{im} x' \Rightarrow \exists a': x'(a') = \beta(b) - b'$

$\exists a: \alpha(a) = a'$. $A, B, B' = A, A', B' \Rightarrow x'(\alpha(a)) = \beta(x(a))$

Letting $b_2 = x(a) \Rightarrow x'(a') = \beta(b) - b' = \beta(b_2)$

$\Rightarrow b' = \beta(b - b_2) \Rightarrow \beta$ is surjective.

8. ($\Rightarrow$) Let $f: M \to N$ be surjective homomorphism and

$g_1 : P_1 \to N$ be arbitrary homomorphism.

Define $g: P_1 \oplus P_2 \to N$ by $g = g_1 \circ \pi_1$ ($\pi_1$ is projection of $P_1$).

$g(x,y) = g_1(x)$.

$\exists h: P_1 \oplus P_2 \to M$: $f \circ h = g$

$h_1 = h \circ i_1$, which commutes

$f \circ h_1 = f \circ (h \circ i_1) = (f \circ h) \circ i_1$

$\Rightarrow f \circ h_1 = g_1$, by symmetry $P_2$ also projective

($\Leftarrow$) Split $g$ into $g_1 = g \circ i_1 : P_1 \to N$

$g_2 = g \circ i_2 : P_2 \to N$

~~(Combine lifts $\Rightarrow$ $h(x,y)$~~ $=$

Lift individually $h_1 : P_1 \to M$ $f \circ h_1 = g_1$

$h_2 : P_2 \to M$ $f \circ h_2 = g_2$

Combine lifts $h(x,y) = h_1(x) + h_2(y)$.

$f(h(x,y)) = g_1(x) + g_2(y) \Rightarrow f \circ h = g$.

9. ($\Rightarrow$) $f: A \to B$ be injective. $g_1 : A \to Q_1$ be homomorphism.

Composition $i_1 \circ g_1 : A \to Q_1 \oplus Q_2$ extends to homomorphism $h: B \to Q_1 \oplus Q_2$, because $Q_1 \oplus Q_2$ injective. Thus, $h \circ f = i_1 \circ g_1$.

Define $h_1 = \pi_1 \circ h : B \to Q_1$.

$h_1 \circ f = \pi_1 \circ (h \circ f) = \pi_1 \circ (i_1 \circ g_1) = (\pi_1 \circ i_1) \circ g_1 = g_1$

($\Leftarrow$). Project $g$: $g_1 = \pi_1 \circ g$ $g_2 = \pi_2 \circ g$,

$Q_1$ and $Q_2$ injective, homomorphism $h_1 : B \to Q_1$ and $h_2 : B \to Q_2$

st $h_1 \circ f = g_1$ $h_2 \circ f = g_2$.

Define $h : B \to Q_1 \oplus Q_2$ by $h(b) = (h_1(b), h_2(b))$.

$\Rightarrow$ Lifts.

10. ($\Rightarrow$) $f \otimes id_{\oplus A_i}$ is injective.

$\Rightarrow \bigoplus_{i \in I} (f \otimes id_{A_i})$ is injective.

Map $f \otimes id_{A_i} : M \otimes A_i \to N \otimes A_i$ is injective

$\Rightarrow A_i$ is flat

($\Leftarrow$) induced map $f \otimes id_{A_i}$ is injective $\forall i \in I$.

Direct sums are injective, ~~so~~ map $\bigoplus_{i \in I} (f \otimes id_{A_i})$.

$\Rightarrow$ injective, $\Rightarrow \bigoplus_{i \in I} A_i$ is flat.

PS 7.

1. a Suppose $\forall M$, $M$ is $R$-module,

(⇒) $M$ is projective.

Consider SES $0 \to M \to A \to B \to 0$, $A, B$ are $R$-modules,

By assumption, this SES splits ⇒ $M$ is direct summand of $A$ ⇒ $M$ is injective.

(⇐) Similarly, consider SES $0 \to A \to B \to M \to 0$,

~~Since~~ By hypothesis, $A$, is injective ⇒ SES splits

⇒ $M$ is direct summand ⇒ $M$ is projective.

2. a Every finite abelian group is direct sum of finite cyclic groups. $A \cong \mathbb{Z}/n_1\mathbb{Z} \oplus \mathbb{Z}/n_2\mathbb{Z} \dots \oplus \mathbb{Z}/n_k\mathbb{Z}$

Consider SES $0 \to n\mathbb{Z} \to \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} \to 0$,

Suppose $\mathbb{Z}/n\mathbb{Z}$ projective ⇒ $\mathbb{Z} \cong n\mathbb{Z} \oplus \mathbb{Z}/n\mathbb{Z}$.

But $n(0,1) = (0,0)$, $(0,1) \neq (0,0)$,

but $0$ is only element $\in \mathbb{Z}$ st $n0 = 0$ ⇒ $\mathbb{Z}/n\mathbb{Z}$ is not projective. ⇒ $A$ is not projective.

b.
$$\begin{tikzcd} n\mathbb{Z} \arrow[r, "i"] \arrow[d, "f"'] & \mathbb{Z} \arrow[dl, dashed] \\ \mathbb{Z}_n & \end{tikzcd}$$

Suppose $\mathbb{Z}_n$ is injective, then for homomorphism $f: n\mathbb{Z} \to \mathbb{Z}_n$, $nx \to x \bmod n$.

$\exists F: \mathbb{Z} \to \mathbb{Z}_n$ ~~nx→~~ st ~~h~~ $F \circ i = f$

⇒ $F(nx) = f(nx)$, since $i$ injective.

We know $F(n) = f(n) = 1$, Also

$F(n) = n \cdot F(1) = 0$ in $\mathbb{Z}_n$ ⇒ contradiction,

~~⇒ A not injective~~ $\mathbb{Z}/n\mathbb{Z}$ not injective ⇒ $A$ not injective

3. a. Let $F_1, F_2$ be free $R$-modules $F_1 \cong \bigoplus_{i \in I} R$, $F_2 \cong \bigoplus_{j \in J} R$

$$F_1 \otimes_R F_2 = \left(\bigoplus_{i \in I} R\right) \otimes \left(\bigoplus_{j \in J} R\right) = \bigoplus_{i \in I} \left(R \otimes \left(\bigoplus_{j \in J} R\right)\right)$$

$$= \bigoplus_{i \in I} \bigoplus_{j \in J} (R \otimes_R R) = \bigoplus_{(i,j) \in I \times J} R$$ ⇒ free $R$-modules

b. Since $P_1, P_2$ projective. $\exists Q_1, Q_2$ st $P_1 \oplus Q_1 \cong F_1$, $P_2 \oplus Q_2 \cong F_2$ ⇒ $F_1 \otimes F_2$ is free $R$ module.

$P_1 \otimes P_2$ projective

$F_1 \otimes F_2 = (P_1 \oplus Q_1) \otimes (P_2 \oplus Q_2) = (P_1 \otimes P_2) \oplus (P_1 \otimes Q_2) \oplus (Q_2 \otimes P_2) \oplus (Q_1 \otimes P_2) \oplus (Q_1 \otimes Q_2)$ ⇒ $P_1 \otimes P_2$ is direct summand of free module

4. a. Let $f: \bigoplus_{i \in I} B_i \to A$. be a homomorphism

$\forall k \in I$, define $f_k : B_k \to A$ as $f_k(b) = f(0, \ldots, \underset{\uparrow k^{th} \text{ pos.}}{b}, \ldots, 0)$

~~The homo.~~ Define $\phi$: Hom $\to \prod$ Hom as

$\phi(f) = (f_k)_{k \in I}$

~~Let inverse map be defined as $\phi^{-1}((f_i)_{i \in I}) = \sum_{i \in I} f_i$~~

Build ~~inverse~~ $f: \bigoplus_{i \in I} B_i \to A$ as $f((b_i)_{i \in I}) = \sum_{i \in I} f_i(b_i)$

This is both left and right inverse.

$= \phi^{-1}((f_i)_{i \in I})$

b. Let $g: A \to \prod_i B_i$. The output $g(a)$ is tuple, let $g_k(a)$ be element at $k^{th}$ position of tuple. $\Rightarrow$ define $\phi$: as

$\phi(g) = (g_k)_{k \in I}$.

Inversely, suppose given $(g_i)_{i \in I}$. $g_i : A \to B_i$. Let $g(a) = (g_i(a))_{i \in I}$ $\Rightarrow$ They are clearly inverses.

5. a. ($\Rightarrow$) ~~im $\psi'$ = ker $\varphi'$~~ $\varphi'$ is surjective.

For $D = N$, $\exists \mu$: ~~D~~ $N \to M$, st $\varphi'(\mu) = id_N$

$\Leftrightarrow$ $\varphi \circ \mu = id$ $\Rightarrow$ original sequence is split.

($\Leftarrow$) $\exists \mu : N \to M$, st $\varphi \circ \mu = id_N$. We also know

$0 \to Hom_R(D, L) \to Hom(D, M) \to Hom(D, N)$ ~~$\to 0$~~ is exact

We must show $\varphi'$ is surjective. For $f \in Hom_R(D, N)$,

Must find $g \in Hom(D, M)$ st $\varphi'(g) = f$, let $g = \mu \circ f$

$\Rightarrow \varphi'(\mu(f)) = f$ $\Rightarrow$ ~~we for~~ $\varphi'$ surjective, ~~so~~ $\square$

b. ($\Rightarrow$) ~~Let $N = D$ again, since $\varphi'$ is injective,~~

~~$\varphi'(id_N) = id_{M,N}$~~

Let $L = D$, now $\exists \lambda : M \to D$: $\psi'(\lambda) = id_L$ since $\psi'$ surjective. $\psi'(\lambda) = \lambda \circ \psi$. So ~~$\lambda$ is~~ the original seq splits.

($\Leftarrow$). We know $0 \to Hom(N, D) \xrightarrow{\varphi'} Hom(M, D) \xrightarrow{\psi'} Hom(L, D)$ is exact.

Must show $\psi'$ is surjective. Let $f \in Hom(L, D)$,

must find $g \in Hom(M, D)$ st ~~$g$~~ $g \circ \psi = f$. $\Leftrightarrow \psi'(g) = f$.

$\exists \lambda : M \to L$: $\lambda \circ \psi = id$. Let $g = f \circ \lambda$, then we get $\psi'(g) = f$, since it splits.

6 Since $P[x]$ is free, it is flat. It is free since it's basis is $\{1, x, x^2, \dots\}$ and is linearly independent.

7. Let $A, B$ be left-$S$ modules. Consider a exact sequence

$0 \to A \xrightarrow{f} B$, where $f$ is injective. Tensoring by $N$,

$\Rightarrow 0 \to N \otimes_S A \xrightarrow{1 \otimes f} N \otimes_S B$, where $N$ is ~~right R-mo.~~ $(R,S)$-bimodule.

$\hookrightarrow$ this is exact since $N$ is flat.

Apply $M$:

$$0 \to M \otimes_R (N \otimes_S A) \xrightarrow{1_M \otimes (1_N \otimes f)} M \otimes_R (N \otimes_S B)$$

this is exact since $M$ is flat

but $M \otimes_R (N \otimes_S A) \cong (M \otimes_R N) \otimes_S A$.

$\hookrightarrow (M \otimes_R N)$ is flat.

8. Since $R$ is commutative, $(R,S)$-bimodule $N \Leftrightarrow R$-module.

$\Rightarrow M \otimes N$ is flat directly.

9. Let subspace be: $W := \{(x_1, \dots, x_n) : a_1x_1 + \dots + a_nx_n = 0\}$.

$(0, 0, \dots, 0) \in W$. Let $u, v \in W$,

$a_1(u_1+v_1) + \dots a_n(u_n+v_n) = 0 + 0 = 0 \Rightarrow u+v \in W$.

~~$\lambda(u+v)$~~ $a_1(\lambda u_1) + \dots a_n(\lambda u_n) = \lambda(a_1u_1 + \dots + a_nu_n) = 0$

$W$ is subspace.

Since $a_1x_1 + \dots + a_nx_n = 0 \Rightarrow x_1 = -\frac{a_2}{a_1}x_2 - \frac{a_3}{a_1}x_3 - \dots - \frac{a_n}{a_1}x_n$

we have $n-1$ choices for $x_i$ which uniquely determine $x_1$. The basis is $v_2 = (-\frac{a_2}{a_1}, 1, 0, \dots)$ $\dots$ $v_n = (-\frac{a_n}{a_1}, 0, \dots, 0, 1)$.

$v_3 = (-\frac{a_3}{a_1}, 0, 1, \dots)$

10. Clearly, $1, x, x^2, \dots, x^5$ span $V$. Also it is closed under addition/scalar multiplication.

Now, ~~suppose~~ we must show $1, 1+x, \dots, 1+x+\dots+x^5$ is linear independent, since there are 6 elements.

~~$c_1 + (c_1+c_2)x$~~ $c_1 + c_2(1+x) + \dots c_6(1+x+\dots+x^5) = 0$

$= (c_1 + c_2 + c_3 + \dots + c_6) + (c_2 + \dots + c_6)x + \dots + c_6x^5 = 0$

$\Rightarrow c_6 = 0 \Rightarrow c_5 = 0 \Rightarrow \dots \Rightarrow c_1 = 0$ $\blacksquare$

PS 8.

1. $\varphi(1) = 1$

$$\begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & +1 & -1 & 0 & 0 \\ 0 & 0 & +1 & -1 & 0 \\ 0 & 0 & 0 & +1 & -1 \\ 0 & 0 & 0 & 0 & +1 \end{bmatrix}$$

since.

$(1+x) - \cancel{1} = x.$

2. $1 \to 1$

$\cancel{x^n}\lambda \quad x \to x - \lambda = 1 \cdot x - \lambda \cdot 1$

$\cancel{x^j} = \quad x = (x-\lambda) + \lambda$

$x^j = \sum \binom{j}{i} (x-\lambda)^i \lambda^{j-i}$

$j^{th}$ col.:

$$\begin{array}{r} \\ \\ i^{th} \text{ row} \\ \\ \\ \end{array} \begin{bmatrix} \lambda^j \\ j\lambda^{j-1} \\ \vdots \\ \binom{j}{i}\lambda^{j-i} \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

3. Note $R_1$ and $R_3$, $R_2$ and $R_4$ proportional.

$\Rightarrow$ Reduce to two row vectors $\Rightarrow$

$$\begin{bmatrix} 1 & -1 & 0 & 3 \\ 1 & 2 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 & 3 \\ 0 & 1 & +1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$ Find $\ker\varphi$.

free variables $x_3, x_4$ $\quad x_2 = -x_3 - 2x_4$

$$\Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} -x_3 - 2x_4 - 3x_4 \\ -x_3 - 2x_4 \\ x_3 \\ x_4 \end{pmatrix} = x_3 \underbrace{\begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \end{pmatrix}}_{\ker\varphi} + x_4 \underbrace{\begin{pmatrix} -5 \\ -2 \\ 0 \\ 1 \end{pmatrix}}_{\ker\varphi.}$$

Basis of $\ker\varphi$.

$$\operatorname{im}\varphi = \operatorname{span}\underbrace{(c_1, c_2, c_3, c_4)}_{\text{columns of } A} = \operatorname{Span}\left\{\begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 2 \\ 1 \\ -2 \end{pmatrix}\right\}$$

4. a. $a_1\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + a_2\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + \dots a_4\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = 0 \Rightarrow a_1 = a_2 = a_3 = a_4 = 0$

Additionally, it spans $V$.

b. $\varphi\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}\right) = \varphi\begin{pmatrix} a+a' & b+b' \\ c+c' & d+d' \end{pmatrix} = a+a'+d+d' = \varphi\begin{pmatrix} a & b \\ c & d \end{pmatrix} + \varphi\begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$

$\varphi(k\begin{pmatrix}a & b\\ c & d\end{pmatrix}) = \varphi\begin{pmatrix}ka & kb\\ kc & kd\end{pmatrix} = ka + kd = c\varphi\begin{pmatrix}a & b\\ c & d\end{pmatrix}$

$\dim V = 4 \Rightarrow \dim \mathbb{R} = 1$

$\varphi\begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix} = 1$, $\varphi\begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix} = 0$ $\varphi\begin{pmatrix}0 & 0\\ 1 & 0\end{pmatrix} = 0$ $\varphi\begin{pmatrix}0 & 0\\ 0 & 1\end{pmatrix} = 1$

$\Rightarrow \mathcal{M}(\varphi) = [1, 0, 0, 1]$

$$\begin{pmatrix}a_1 & a_2\\ a_3 & -a_1\end{pmatrix} = \underbrace{a_1\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix} + a_2\begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix} + a_3\begin{pmatrix}0 & 0\\ 1 & 0\end{pmatrix}}_{\ker\varphi}$$

$\Rightarrow$

$\dim \ker\varphi = 3.$

5. a) $f = a_0 + a_1x + \dots + a_6x^6$, $\varphi(f) = \cancel{a_1 + a_2x + \dots}\ a_1 + 2a_2x + \dots + 6a_6x^5 = 0$

$\ker\varphi = \cancel{\{a_0 \in \mathbb{R}\}}\ \{a_0 \mid a_0 \in \mathbb{R}\} \Rightarrow$ basis is $\{1\}$

$\operatorname{im}\varphi$: basis is $\{1, x, \dots, x^5\}$

b). $\ker\varphi$: basis is ~~even~~ polynomials who's coefficients of an odd power are zero $\{x^2 + x^4 + x^6 + a_0\} \Rightarrow \{1, x^2, x^4, x^6\}$.

$\operatorname{im}\varphi$: polynomials of form ~~6x~~ $5x^4 + 3x^2 + 1$

basis is $\{1, x^2, x^4\}$

c). $\ker\varphi$: $\cancel{a_1 + 2a_2x + 4a_4x^3 + 5a_5x^4} \Rightarrow a_3x^3 + a_6x^6 + a_0$

$\Rightarrow$ basis $= \{1, x^3, x^6\}$.

$\operatorname{im}\varphi$: $a_1 + 2a_2x + 4a_4x^3 + 5a_5x^4 \Rightarrow \{1, x, x^3, x^4\}$.

d). $\ker\varphi$: $a_0 + a_5x^5 \Rightarrow \{1, x^5\}$.

$\operatorname{im}\varphi$: $a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + 6a_6x^5 \Rightarrow \{1, x, x^2, x^3, x^5\}$.

6. a. $E(p) = p(3)$ $E(cp+q) = (cp+q)(3) = cp(3) + q(3) = cE(p) + E(q)$

$\Rightarrow E \in V^*$ ~~Basis~~ $E(x^i) = 3^i$

$\Rightarrow E = v_0^* + 3v_1^* + 9v_2^* + 27v_3^* + 81v_4^* + 243v_5^*$.

b. $\varphi(p) = \int_0^1 p\,dt$ $\varphi(cp+q) = \int_0^1 cp+q\,dt = c\varphi(p) + \varphi(q) \Rightarrow \varphi \in \mathrm{Hom}(V; \mathbb{R})$.

$\cancel{\varphi(2t)} = \varphi(x^i) = \frac{1}{i+1} \Rightarrow \varphi = 1 \cdot v_0^* + \frac{1}{2}v_1^* + \frac{1}{3}v_2^* + \frac{1}{4}v_3^* + \frac{1}{5}v_4^* + \frac{1}{6}v_5^*$

c. Similarly true proof to (b)

$\varphi(x^i) = \int_0^1 t^{i+2}\,dt = \frac{1}{i+3} \Rightarrow \varphi = \frac{1}{3}v_0^* + \frac{1}{4}v_1^* + \frac{1}{5}v_2^* + \dots + \frac{1}{8}v_5^*$

d. $\varphi(cp+q) = cp'(5) + q'(5) = c\varphi(p) + \varphi(q) \Rightarrow \in V^*$.

$\varphi(x^i) = i5^{i-1} \Rightarrow \varphi = 0v_0^* + 1v_1^* + 10v_2^* + 75v_3^* + 500v_4^* + 3125v_5^*$

7. ($\Rightarrow$) columns = basis $\Rightarrow$ invertible $\Rightarrow$ $\det A \neq 0$

($\Leftarrow$) $\det \neq 0 \Rightarrow$ invertible $\Rightarrow$ $Ax=0$ has only trivial solution $\Rightarrow$ columns are linearly independent

8 $Sym_2(x\otimes y) = x\otimes y + y\otimes x$ since only two possible permutations.

but ~~xy~~ $x\otimes y = xy(1\otimes 1) \Rightarrow Sym_2(x\otimes y) = 2xy(1\otimes 1)$.

$a = 2xy$ even. If $1\otimes 1 \in \operatorname{im} Sym_2 \Rightarrow$ impossible.

9. $\Lambda^k(M) = \underbrace{M\otimes \dots \otimes M}_{i \text{ times}}$, but we don't allow repeats and ordering doesn't matter since $m\wedge n = -n\wedge m$ $\Rightarrow \binom{n}{i}$. This set of $\binom{n}{i}$ elements from spanning set of $M$ spans $\Lambda^i(M)$. It is also linearly independent by universal property.

Factor $f_J : \Lambda^i \to R$ $\quad f_J(e_J) = 1 \quad f_J(e_I) = 0$ $\quad I \neq J$

Apply $f_J$ to linear combination to show coefficients $= 0$

10. Show $M$ is $R$-module.

$$\sigma(\sigma(e_1)) = \sigma(e_1 + 2e_2) = e_1 + 2e_2 + 2\sigma(e_2) = e_1$$

$\sigma(\sigma(e_2)) = e_2$ ✓ $\quad \sigma^2 = $ identity.

$$\sigma(e_1\wedge e_2) = (e_1 + 2e_2)\wedge e_2 = e_1\wedge e_2 + 2(\cancel{e_2\wedge e_2}) = e_1\wedge e_2$$

$= 0$

Similarly, $\sigma(e_1\wedge e_2) = e_1\wedge(-e_2) = -(e_1\wedge e_2)$

$\Rightarrow 2(e_1\wedge e_2) = 0 \Rightarrow$ order 2.

$(e_1\wedge e_2)$ can't be 0.